

Cassini ISS F Ring Mosaics User Guide

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Versions and errata. This is Version 1.0 of the bundle and of this *User Guide*. The bundle version is recorded in the `version_id` of every label, and each label carries a `Modification_History` describing what changed in each version. Corrections made after release are issued as a new version of the bundle, with a new `Modification_Detail` entry recording the change and a reissue of this *User Guide* at the matching version. Errata identified between releases are posted with the bundle at the PDS Ring-Moon Systems Node (<https://pds-rings.seti.org/>). Please report suspected errors to rfrench@seti.org.

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1 Introduction

Saturn's F ring is one of the most dynamic and scientifically intriguing features in the solar system. Discovered in 1979 by the Pioneer 11 spacecraft (Gehrels et al. 1980), the F ring is a narrow, complex, and ever-changing structure located just outside Saturn's main ring system. Its unique morphology, characterized by multiple strands, clumps, and transient features, along with the influence of neighboring moons Prometheus and Pandora, has made it a focal point for studies of planetary ring dynamics, satellite-ring interactions, and the processes governing the evolution of ring systems.

Before 2004, good-quality images of the F ring were limited to a handful produced by the Voyager 1 and 2 spacecraft during their short flybys in 1980 and 1981. This all changed with the arrival of the Cassini Orbiter at Saturn on July 1, 2004. During its subsequent 13 years at Saturn, Cassini used its Imaging Science Subsystem (ISS) (Knowles 2018; Porco et al. 2004) to observe the F ring more than 100,000 times, providing a treasure trove of data that is unlikely to be surpassed for generations.

This *User Guide* provides a detailed description of the organization and contents of the Cassini ISS F Ring Mosaics bundle archived at NASA's Planetary Data System Ring-Moon Systems Node under a NASA Cassini Data Analysis Program (CDAP) grant, 80NSSC21K0527: "The recent history of Saturn's dusty rings". The purpose of this archive is to provide a carefully curated and documented set of high-quality images of the F ring taken by the ISS Narrow Angle Camera (NAC) and Wide Angle Camera (WAC) (Porco et al. 2004) using clear filters. Special emphasis is placed on F movies, observations whose names contain "FMOVIE": sequences of dozens to hundreds of images taken in rapid succession at a single inertial longitude while the F ring orbited through the field of view. Such movies provide a detailed picture of the F ring material at various corotating longitudes as it passes through some true anomaly in the F ring's eccentric orbit.

All included images were calibrated in units of I/F using CISSCAL 4.0 (Knowles 2018) and then reprojected onto a grid with radius relative to the F ring core on the vertical axis and corotating longitude on the horizontal axis (see Section 3.4). Multiple images were stitched together to form more complete mosaics, and the mosaics were then analyzed to remove any background gradient.

The archive includes reprojected images, mosaics, and mosaics that have been adjusted by subtracting a background model, organized by Cassini "observation name" (see Section 3.1). Each reprojected image and mosaic is supplemented by a table containing metadata such as resolution, phase angle, and emission angle for each valid corotating longitude. Reprojected images have an additional file describing the spacecraft pointing used to perform the reprojection. Mosaics have an additional file listing the reprojected images that were used to create the mosaic. A series of browse images in various sizes is provided for each reprojected image and mosaic as well. In total, 20,584 reprojected images are provided along with 305 associated mosaics. Of the reprojected images, 20,441 contributed data to a mosaic; the remaining 143 come from observations that archive every image whether or not it was used (Section 3.1.6).

Finally, three global index files are provided covering the complete set of reprojected images, mosaics, and background-subtracted mosaics. These index files summarize metadata such as time, corotating longitude coverage, lighting and viewing geometry, and orbit details for each data product, making it easy to search for and analyze the data products *en masse*.

First-time users are encouraged to read or skim this entire *User Guide* at least once (and possibly twice) before making use of the archived results.

2 Quick start

The dataset described here is archived using the PDS4 standard. While deep knowledge of PDS4 is not required to make use of the data, some familiarity is useful. Specifications, tools, and training are available from the main PDS website at <https://pds.nasa.gov>.¹

This data is archived as a single PDS4 *bundle*, which consists of multiple *collections*. Each collection represents one particular type of data (reprojected images, mosaics, browse images, etc.). Within a collection, individual data products are described by a *label* (extension `.lblx`), which contains substantial information about the data product, including details of any associated binary image (`.img`), graphical image (`.png`), metadata table (`.tab`), or supplemental (`.txt`) files.

Data product files are best read by first processing the associated label file, extracting the needed metadata about the data product, and then using the information in the label to identify the file(s) to read and the details of their format. Special software is available in a variety of programming languages for processing labels and reading the associated data products (Section 5).

A few abbreviations and naming conventions are used throughout this *User Guide*:

- **OBSID** — a Cassini observation name converted to lowercase, including any numeric chunk suffix; it names a mosaic and the directory that holds it (Sections 3.1 and 4.2).
- **IMGID** — the identifier of a single Cassini ISS source image: its ten-digit spacecraft clock (SCLK) count followed by *n* for the NAC or *w* for the WAC, as in 1622049830*n* (Section 4.2).
- **LID** — a PDS4 *logical identifier*, the permanent name of a product, collection, or bundle, written as a `urn:nasa:pds: URN`.
- **LIDVID** — a LID with a version appended after `::`, naming one specific version of that product.
- **I/F** — the calibrated radiometric unit of the image data: the ratio of observed intensity to the incident solar flux, for a perfect Lambert surface at normal incidence (Section 3.2).
- **COISS** — the PDS holdings of raw and calibrated Cassini ISS images, from which the source images for this bundle were drawn (Section 3.2).

Below is a summary of the collections related to reprojected images and the files contained within each.

Table 1: Collections and files for reprojected images.

Collection	Filename(s)	Description
browse_reproj_img (Section 4.2.2)	IMGID_browse_reproj_img.lblx	PDS4 label describing the four types of browse images for reprojected image data.
	IMGID_browse_reproj_img_full.png IMGID_browse_reproj_img_med.png IMGID_browse_reproj_img_small.png IMGID_browse_reproj_img_thumb.png	Human-viewable versions of reprojected images in four sizes in standard PNG format. Images are contrast-stretched and not calibrated for scientific use.
data_reproj_img (Section 4.2.1)	IMGID_reproj_img.lblx	PDS4 label describing the reprojected image and the three associated data files.

¹Refer to the documents in Section 6.1 for further information on the PDS4 concepts of *bundle*, *collection*, and *data product*.

Collection	Filename(s)	Description
	IMGID_reproj_img.img	Binary file containing a $401 \times N$ array of 32-bit floating-point values calibrated as I/F: 401 rows of radius in increments of 5 km, by N columns of corotating longitude in 0.02° increments. The longitudes are restricted to the range containing valid data.
	IMGID_reproj_img_metadata_params.tab	Fixed-width comma-delimited table containing metadata for each valid corotating longitude in the reproj_img.
	IMGID_reproj_img_suppl.txt	Text file containing a description of the navigation results for the original Cassini ISS image and the resulting SPICE C matrix.

Below is a summary of the collections related to mosaics (and background-subtracted mosaics) and the files contained within each.

Table 2: Collections and files for mosaics and background-subtracted mosaics.

Collection	Filename(s)	Description
browse_mosaic [original] browse_mosaic_bkg_sub [background gradient removed] (Section 4.2.4)	OBSID_browse_mosaic[_bkg_sub]. lblx	PDS4 label describing the four types of browse images for mosaic data.
	OBSID_browse_mosaic[_bkg_sub]_full.png OBSID_browse_mosaic[_bkg_sub]_med.png OBSID_browse_mosaic[_bkg_sub]_small.png OBSID_browse_mosaic[_bkg_sub]_thumb.png	Human-viewable versions of mosaics in four sizes in standard PNG format. Images are contrast-stretched and not calibrated for scientific use.
data_mosaic [original] data_mosaic_bkg_sub [background gradient removed] (Section 4.2.3)	OBSID_mosaic[_bkg_sub].lblx	PDS4 label describing the mosaic and the three associated data files.
	OBSID_mosaic[_bkg_sub].img	Binary file containing a $401 \times 18,000$ array of 32-bit floating-point values calibrated as I/F: 401 rows of radius in increments of 5 km, by 18,000 columns of corotating longitude in 0.02° increments.
	OBSID_mosaic[_bkg_sub]_metadata_params.tab	Fixed-width comma-delimited table containing metadata for each valid corotating longitude in the mosaic.

Collection	Filename(s)	Description
	OBSID_mosaic[_bkg_sub]_metadata_src_imgs.tab	Fixed-width comma-delimited table containing a map of image numbers from the metadata_params table to the associated reprojected image LIDVID.

Other collections are provided in the bundle, including a document collection containing this *User Guide* and a miscellaneous collection containing global index files. All collections and support files are described in the sections below.

The remainder of this *User Guide* provides detailed guidance on the use of the bundle:

- Section 3 describes the selection and processing of Cassini ISS images and the creation of the mosaics, including the steps:
 - Image selection (Section 3.1)
 - Image calibration (Section 3.2)
 - Image navigation (pointing correction) (Section 3.3)
 - Image reprojection (Section 3.4)
 - Mosaic creation (Section 3.5)
 - Mosaic background gradient modeling and subtraction (Section 3.6)
- Section 4 describes the bundle organization and directory structure, including the details of all files included
- Section 5 discusses techniques for reading label and data product files
- Section 6 contains useful external references
- Section 7 provides an itemized list of all metadata fields and their meaning

3 Image selection and processing

3.1 Image selection

Cassini observations were organized into observing campaigns. Each campaign was given a unique identifier (the observation name), which has the general format:

```
[INST]_[REV] [TI]_[UNIQUENAME]_[PINST]
```

where for the observations included in this bundle:

- INST is the abbreviation for the instrument being commanded (usually “ISS”)
- REV is the rev (orbit) identifier, starting with 000 at Saturn orbit insertion and ending with 293 at the end of the mission. A few early revs are lettered instead of numbered; this bundle contains rev 00A. Software that splits an observation name should treat REV as a fixed field of three characters and not assume it is an integer.
- TI is a 2-character target identifier: RA (A ring), RB (B ring), RF (F ring), RI (general rings), ST (star)
- UNIQUENAME is a unique descriptor chosen by the observation creator, often containing acronyms such as “LP” for low phase, “HR” for high resolution, and “MOV” for movie. Some of the more common names:
 - APOMOS* – Images taken at Cassini apoapse, often used to make large mosaics
 - AZSCN* – Azimuthal scan around the rings
 - *OCC – Stellar occultation
 - COMPLIT* – Radial composition scan on the lit side of the rings
 - EGAPMOVMP – Encke gap movie, medium phase
 - FMONITOR – F ring monitor; often an ISS rider on CIRS scans of the A and B rings
 - FMOVIE – Stared at a fixed inertial longitude to image F ring material passing through the field of view
 - FMOVIEEQX – F ring movie at Saturn equinox
 - FRSTRCHAN – Prometheus and F ring streamer-channel movie; tracks Prometheus and follows a small section of the F ring around one complete orbit
 - HIRESAFRG – High-resolution A ring outer edge
 - HIRESFRNG – High-resolution F ring
 - LATPHASE – Latitude and phase angle; wide range of sub-spacecraft latitude and solar phase angle
 - PROPRETRG – Re-targeting known “propeller” moons to track their orbits
 - SPK*, SPOKEMOV* – Spoke observation; stared at a ring ansa at regular intervals to watch B ring spokes move through the field of view
 - SUBM*, TDIFS*, TMAP* – Thermal observations by the Composite Infrared Spectrometer (CIRS); ISS rider
 - UR* – Ultraviolet Imaging Spectrograph (UVIS) ring stellar occultation; ISS rider
- PINST usually names the prime instrument when it is not the ISS (Knowles 2018). A value of “PRIME” means the ISS was itself the prime instrument; any other value normally means the ISS

images were a ride-along on an observation designed for the named instrument. The rule is not absolute: alongside “PRIME”, “VIMS”, “CIRS”, and “UVIS”, this bundle also contains the codes “SI” (IOSIC_276RB_COMPLITB3001_SI and IOSIC_276RB_COMPLITB4001_SI) and “PIE” (ISS_253RI_HIRESAFRG001_PIE), which do not name any instrument.

The complete set of $\sim 100,000$ images containing the F ring was manually inspected for high-quality images.

In general, an image was included in our dataset if:

1. It was a member of a series of images taken at different corotating longitudes that could be stitched together into a mosaic;
2. It was taken with the “clear” filters;
3. It was not corrupted or partially downloaded; and
4. Its best radial resolution at the F ring was no coarser than ~ 80 km/pixel.

However, due to the variation in image attributes and quality, the final determination of which images to include was necessarily subjective. For example:

- Some images were included that do not meet requirement #1. If a single WAC image contained enough of the F ring to subjectively make a good mosaic by itself, it was included. Also, some image sequences were included where the camera followed a single corotating longitude as it traversed different inertial longitudes.
- Some images were included that do not meet requirement #4. These included the 42 “SPOKEMOV” mosaics taken at relatively low resolution by the WAC. While the reprojected images for these sequences are not of very high quality, the resulting mosaics are nevertheless sufficient for research, such as photometry, that does not require high-resolution analysis.

In total, 20,584 images were chosen for inclusion.

Image sequences were grouped under their original observation name converted to lowercase. In some cases, a single observation contained multiple distinct image sequences that would make different mosaics. In these cases, a unique numeric suffix was appended to the observation name.

The selected observations were divided into the classes described in the following sections. Every class except the default is indicated in the data archive by a comment in the mosaic PDS4 label and by a note in the global index file; Section 4.4.1 lists the full set of notes.

3.1.1 Standard inertial F movies

A standard “F movie” consists of a series of images taken while Cassini stared at approximately the same inertial longitude. The F ring rotated under the camera for some or all of a complete orbit, providing a partial or complete view of the entire longitudinal extent of the ring. Such observations are considered the default and are thus not marked with a special note.

It is important to realize that it was the F ring that was rotating, not the camera that was instantaneously observing different parts of the ring. The resulting image sequence and mosaic therefore do not provide a view of the F ring around its entire orbit. Instead, they provide repeated views of the F ring centered at (approximately) a single inertial longitude, and thus at a single true anomaly in its eccentric orbit. This can affect the interpretation of morphological features such as streamer-channels, which change shape throughout their orbit (Attree et al. 2012, 2014; Beurle et al. 2010; French et al. 2014a; Murray et al. 2005, 2008).

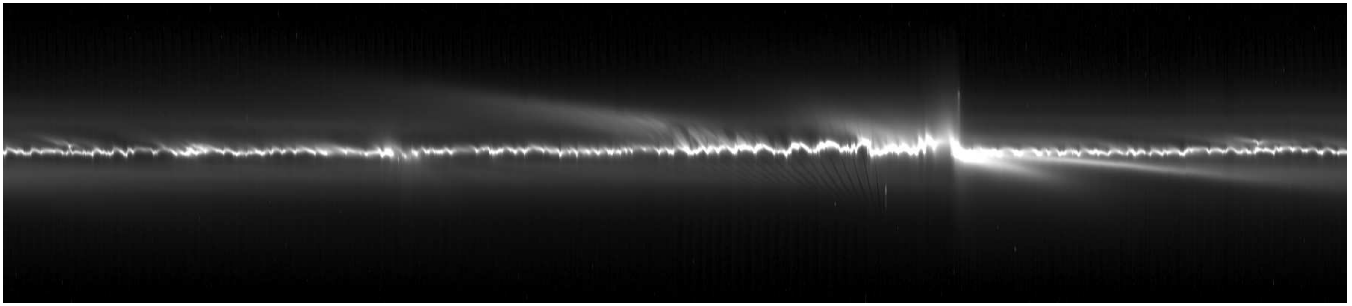


Figure 1: Mosaic of observation *iss_039rf_fmovie001_vims*, a standard *F* movie taken as a rider on a VIMS-prime observation.

An example of a standard mosaic is shown in Figure 1.

3.1.2 *M#*: Observations that are broken into chunks

Although Cassini observing campaigns are indicated by their observation name, sometimes a single campaign produced data that was more appropriately divided into multiple groups. In these cases, we produced multiple mosaics named after the observation name with a numeric suffix. The notes field in the index file (Section 4.4.1) includes “M1”, “M2”, “M3”, or “M4”, indicating how the different chunks are related to each other.

3.1.2.1 *M1*: Multiple contiguous observations of the same inertial longitude range

In some cases, an observation stared at a single inertial longitude for more than a complete orbit of the F ring. In these cases, the observation was broken into multiple chunks, with each complete orbit of the F ring (and the final partial orbit, if there was one) used to create a separate mosaic. Such observations are noted in the index file with “M1”. The following mosaics have this flag. Two shorthands are used in the mosaic lists throughout this section: a suffix list such as *_1/2* stands for the separate mosaics *_1* and *_2*, and a suffix range, written with an en dash outside the name as in *_2–_9*, stands for *_2* through *_9*.

- *iss_198ri_spokemov005_prime_1/2*
- *iss_199ri_spokemov002_prime_1/2*
- *iss_199ri_spokemov003_prime_1/2/3*
- *iss_199ri_spokemov004_prime_1/2*
- *iss_199ri_spokemov006_prime_1/2/3*
- *iss_200ri_spokemov004_prime_1/2/3*
- *iss_201ri_spokemov001_prime_1/2*
- *iss_201ri_spokemov009_prime_1/2*

3.1.2.2 *M2*: Pairs of observations taken at inertial longitudes 180° apart

Some observations were performed in pairs, where a particular range of corotating longitudes was observed at one inertial longitude, and then approximately the same range of corotating longitudes was observed at the inertial longitude roughly 180° away. Such pairs provide insight into how particles at the same corotating longitude are actually on slightly different orbits. These observations are divided into two chunks and noted with “M2”. The following mosaics have this flag:

- *iss_085rf_fmovie003_prime_1/2*

- iss_094rf_fmovie001_prime_1/2
- iss_112rf_fmovie002_prime_1/2
- iss_173rf_fmovie001_prime_1/2
- iss_174rf_frstrchan001_prime_1/2
- iss_177rf_fmovie001_prime_1/2
- iss_177rf_frstrchan001_prime_1/2
- iss_183rf_fmovie001_prime_1/2
- iss_193rf_fmovie001_prime_1/2
- iss_194rf_fmovie001_prime_1/2
- iss_198rf_fmovie001_prime_1/2
- iss_201rf_fmovie001_prime_1/2
- iss_202rf_fmovie001_prime_1/2
- iss_203rf_fmovie001_prime_1/2
- iss_208rf_fmovie001_prime_1/2
- iss_208rf_fmovie002_prime_1/2
- iss_209rf_fmovie001_prime_1/2
- iss_212rf_fmovie001_prime_1/2
- iss_232ri_egapmovmp001_prime_1/2
- iss_233rf_fmovie001_prime_1/2
- iss_235rf_fmovie002_prime_1/2
- iss_239rf_fmovie002_prime_1/2
- iss_243rf_fmovie001_prime_1/2
- iss_256rf_fmovie001_prime_1/2
- iss_276ri_hiresafrg001_prime_1/2

An example of such a pair of mosaics is shown in Figure 2.

3.1.2.3 M3: Multiple observations of the same corotating longitude range but different inertial longitude ranges

While standard “F movies” are taken by staring at a single inertial longitude in isolation, some observations were performed by taking multiple “F movies” in a row, each covering roughly the same range of corotating longitudes but at different inertial longitudes. These observations were split apart and are marked with “M3”. The following mosaics have this flag:

- iss_105ri_tmapn45lp001_cirs_1-_6
- iss_262rf_fmovie001_prime_01-_17
- iss_268rf_fmovie001_prime_2-_9

An example is shown in Figure 3.

3.1.2.4 M4: Observations of different corotating and different inertial longitudes

Finally, some observations follow neither a single corotating longitude nor a single inertial longitude. This is often the case for occultations. If both the ingress and egress occultations are imaged, the star’s position

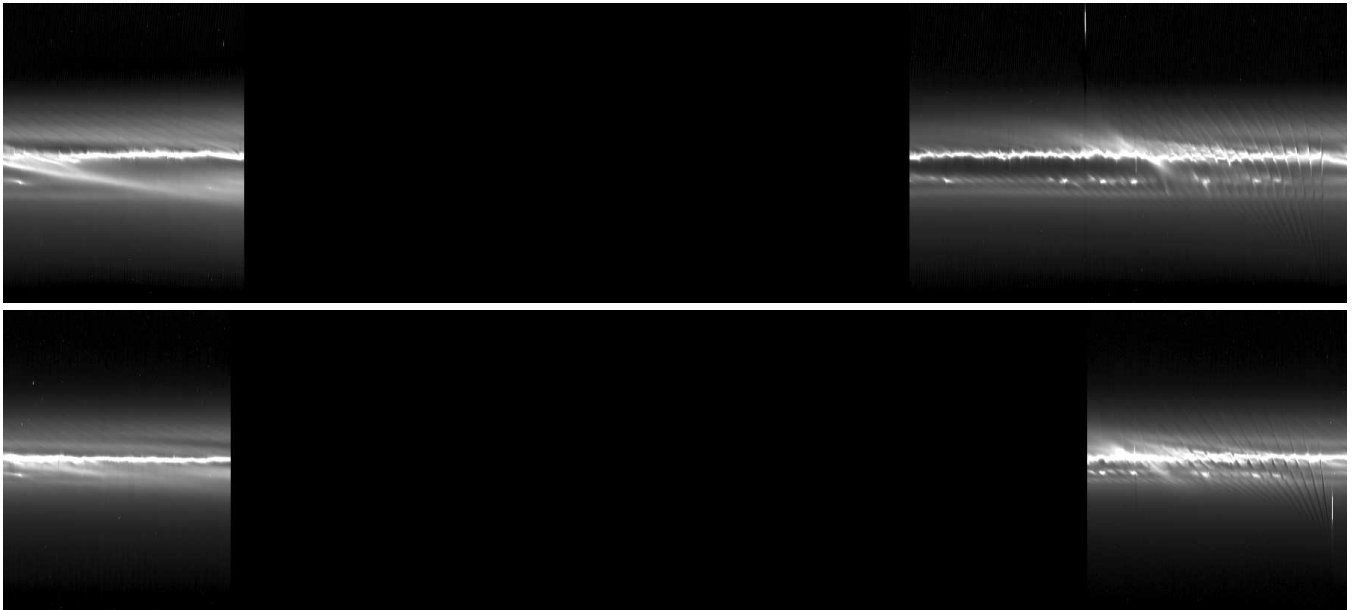


Figure 2: Mosaics of observations *iss_112rf_fmovie002_prime_1* (top) and *iss_112rf_fmovie002_prime_2* (bottom). They cover approximately the same range of corotating longitudes but were taken at inertial longitudes roughly 180° apart. Morphological differences between the two are visible, giving potential insight into the orbits of the particles in these regions. (Flag M2.)

during ingress bears no fixed relation to its position during egress, nor to either the corotating or the inertial longitude. There are also some observations that are broken into different segments similar to “M3”, but each chunk observed a different range of both corotating and inertial longitude. These observations are marked with “M4”. The following mosaics have this flag:

- *iss_185ri_rhyaocc001_vims_1/2*
- *iss_191rf_fmovie001_prime_1/2/3/4/5*
- *iss_201ri_l2pupocc001_vims_1/2*
- *iss_206ri_propretrg001_prime_1/2/3*
- *iss_239ri_hiresafrg001_prime_1/2*
- *iss_253rf_fmovie001_prime_1/2/3*
- *iss_268rf_fmovie001_prime_1* (for this mosaic, all of the other chunks are marked with “M3”, but this chunk behaves differently)

3.1.3 R: Follows one corotating longitude range with different inertial longitudes

A small number of observations follow a single corotating longitude, taking one image at each inertial longitude. These are similar to the “M3” observations above, but contain such a small range of corotating longitudes in each image that it is not worth breaking them into separate mosaics. In these cases, a single mosaic is produced, but because the images overlap *the mosaic is not particularly useful*. It is recommended instead to look at the corresponding reprojected images. These observations are marked with “R”. If a single observation is split into chunks, “R” may be paired with an appropriate “M” note. An example is shown in Figure 4. The following mosaics have this flag:

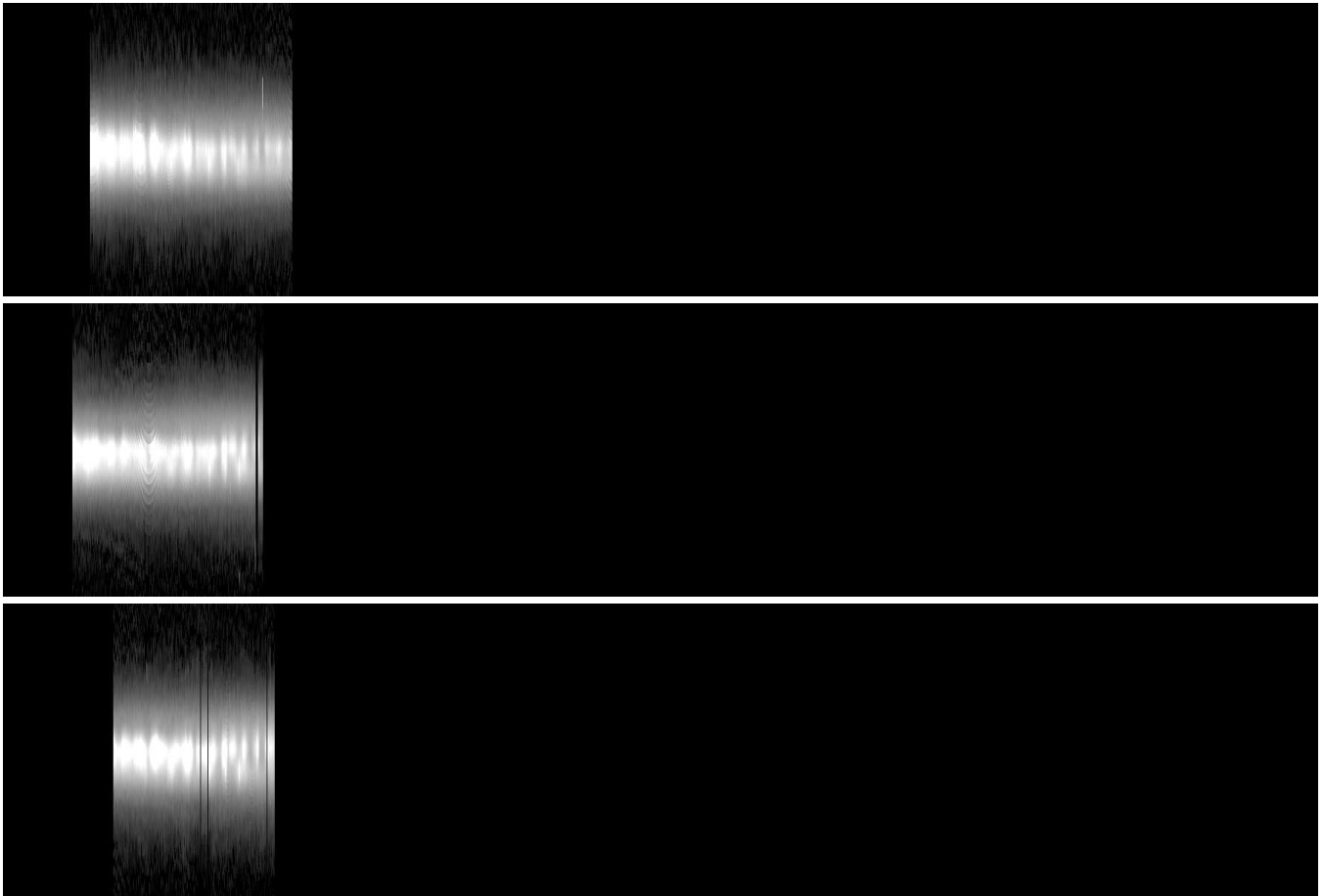


Figure 3: A series of observations covering the same corotating longitudes but taken at successively increasing inertial longitudes. `iss_105ri_tmapn45lp001_cirs_1` (top) was taken at 79° – 131° ; `iss_105ri_tmapn45lp001_cirs_2` (middle) at 109° – 155° ; `iss_105ri_tmapn45lp001_cirs_3` (bottom) at 136° – 186° . (Flag M3.)

- `iss_191rf_fmovie001_prime_1/2/3/4/5`
- `iss_199rf_fmovie002_prime`
- `iss_256ri_hiresafrg002_prime`
- `iss_262rf_fmovie001_prime_12`
- `iss_268rf_fmovie001_prime_1`

3.1.4 N: Neither inertial nor corotating

In a few cases, especially those involving wide-field views taken by the WAC, an observation cannot be classified as either inertial or corotating because the camera does not follow anything; it takes a single image, or a short series of images, covering large portions of the F ring. These cases are marked with “N”. An example is shown in Figure 5. The following mosaics have this flag:

- `iss_007ri_azscnloph001_prime`
- `iss_039ri_fmonitor002_cirs`

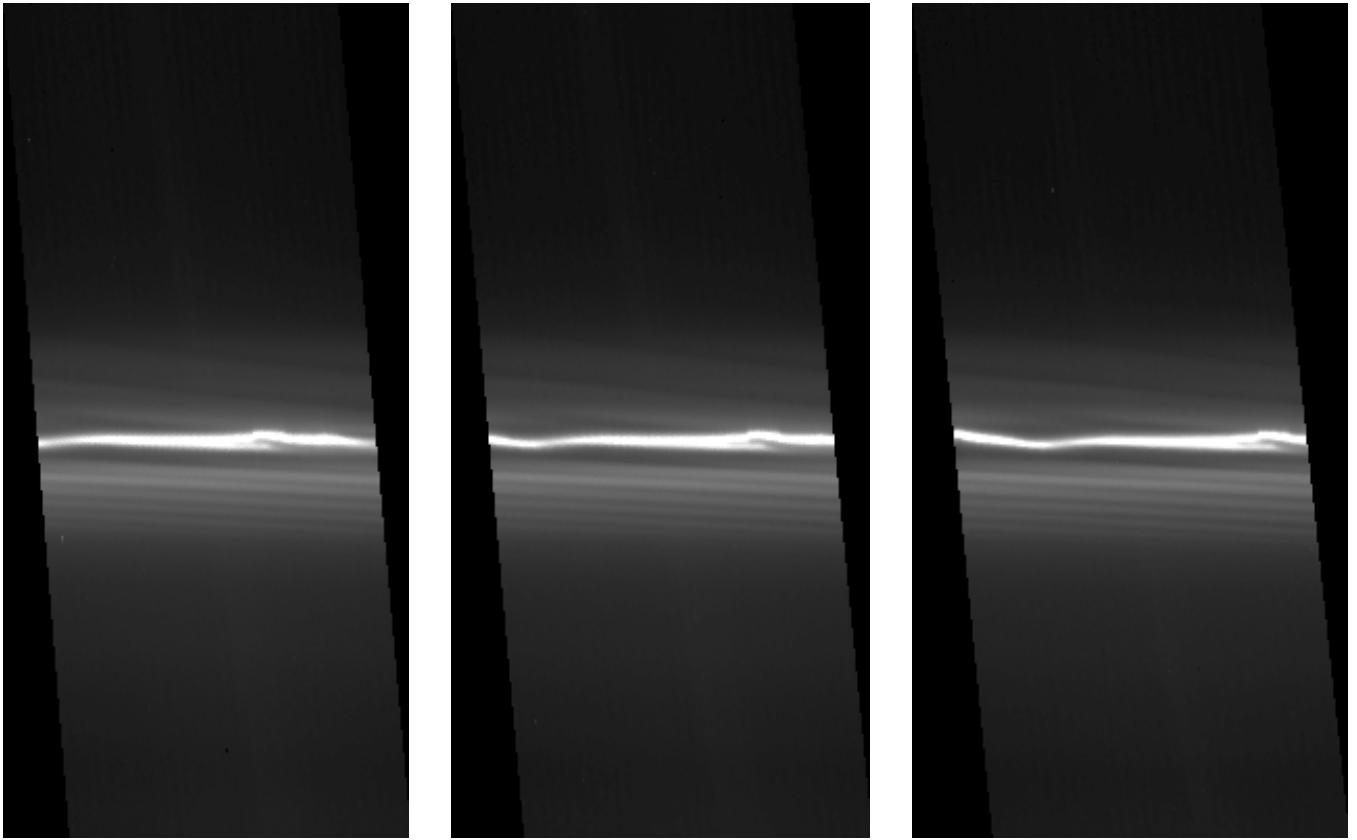


Figure 4: Three reprojected images from mosaic *iss_191rf_fmovie001_prime_1*, each covering approximately the same 4° range of corotating longitude. Source image index 0, 1748745661n (left), was taken at inertial longitudes 5.7° – 9.6° ; index 4, 1748746261n (middle), at 9.3° – 13.3° ; index 8, 1748746865n (right), at 12.9° – 17.1° . (Flag R.)

- *iss_041ri_fmonitor001_cirs*
- *iss_043ri_fmonitor001_cirs*
- *iss_043ri_fmonitor001_prime*
- *iss_079ri_fmonitor002_prime*
- *iss_081ri_fmonitor001_prime*
- *iss_082ri_fmonitor003_prime*
- *iss_083ri_fmonitor002_prime*
- *iss_084ri_fmonitor002_prime*
- *iss_098ri_tmapn30lp001_cirs*
- *iss_100ri_subms20lp001_cirs*

3.1.5 O: Occultation

Several observations were designed to monitor a stellar occultation of the F ring core. These observations continuously observed the position of the star as the F ring rotated. Since each image has the star in it, the mosaic may show the star multiple times. Like the “R” mosaics, these mosaics are *not particularly useful*

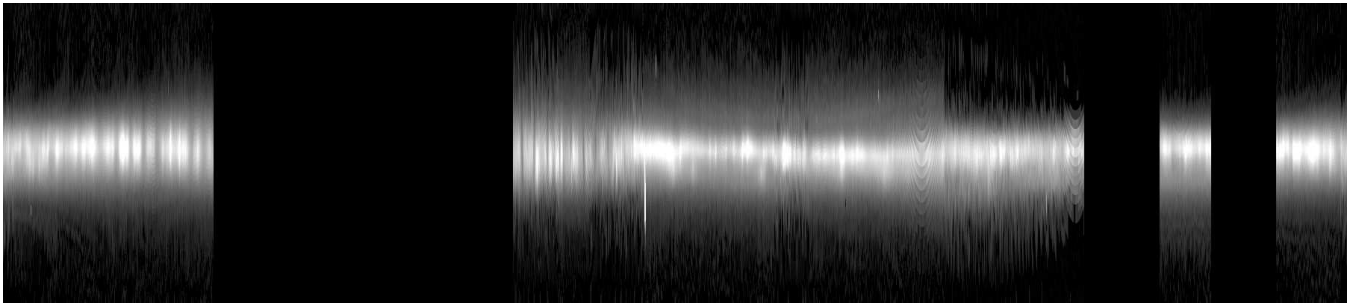


Figure 5: Non-inertial, non-corotating mosaic `iss_098ri_tmapn30lp001_cirs`, which covers most of the full corotating range 0° – 359.98° , taken from inertial longitudes 93° – 310° . (Flag N.)

by themselves and it is recommended to look at the original reprojected images instead. Also, since these are “ride-along” observations where the Cassini Visible and Infrared Mapping Spectrometer (VIMS) or UVIS was the prime instrument measuring the photon flux, the ISS image sequences do not necessarily show the moment when the star was occulted by the F ring.

The observations `iss_185ri_rhyaocc001_vims` and `iss_201ri_l2pupocc001_vims` observed both the ingress and egress and thus are split into two chunks and also marked with “M4”.

The occultation observations are marked with “O”. The following mosaics have this flag, along with the star that each observation was pointed at:

- `iss_172ri_betpegocc001_vims` — Scheat (Beta Pegasi)
- `iss_172st_urgampeg001_uvis` — Algenib (Gamma Pegasi)
- `iss_180ri_rlyrocc001_vims` — R Lyrae (13 Lyrae)
- `iss_180ri_rcasocc001_vims` — R Cassiopeiae
- `iss_185ri_rhyaocc001_vims_1/2` — R Hydrae
- `iss_191ri_rcasoccb001_vims` — R Cassiopeiae
- `iss_194ri_mucepocc001_vims` — Herschel’s Garnet Star (Mu Cephei)
- `iss_196ri_betandocc001_vims` — Mirach (Beta Andromedae)
- `iss_197ri_whyaocc001_vims` — W Hydrae
- `iss_198ri_rlyrocc001_vims` — R Lyrae (13 Lyrae)
- `iss_201ri_l2pupocc001_vims_1/2` — L2 Puppis
- `iss_205ri_l2pupocc002_vims` — L2 Puppis
- `iss_206ri_l2pupocc002_vims` — L2 Puppis

Each of these stars is named in a `Target_Identification` block in the label of every data product belonging to its observation, and is linked there to the PDS context product for the star. The full set of stars observed is listed in `bundle.lblx` and in the three `data_*` collection labels. The `browse`, `document`, `miscellaneous`, `context`, and `xml_schema` collection labels do not name any target at all, and `collection_spice_kernels.lblx` lists only the two ring targets.

3.1.6 Reprojected images that were not used in a mosaic

For most observations, the reprojected images archived here are exactly those that contributed data to the observation’s mosaic. The “R” (Section 3.1.3) and “O” (Section 3.1.5) observations are the exceptions.

For these, the mosaic is of limited use and the reprojected images are the product of interest, so every reprojected image from the observation is archived, whether or not it supplied any data to the mosaic.

A reprojected image that did not contribute to the mosaic is not listed in the mosaic's `OBSID_mosaic[_bkg_sub]_metadata_src_imgs.tab` file and is never referenced by the `image_index` column of `OBSID_mosaic[_bkg_sub]_metadata_params.tab` (Section 4.2.3). The description in each reprojected image's own label states which of the two cases applies. The label of an image that did not contribute does not carry a derived-product reference to the mosaic or to the background-subtracted mosaic, because neither holds any of its pixels. A related case arises when an image did contribute to the mosaic but every longitude it supplied was later removed by the background subtraction (Section 3.6): its label then references the mosaic but not the background-subtracted mosaic, and the background-subtracted mosaic's source-image table omits it (Section 4.2.3).

3.2 Calibration

Calibrated images were retrieved from the PDS Ring-Moon Systems Node archive.² Each image was calibrated by the Node using the CISSCAL 4.0 pipeline (Knowles 2018) into units of I/F.³ At the time of writing, the calibrated files are being migrated to the PDS4 standard, and they will be archived as collections within the `cassini_iss_cruise` and `cassini_iss_saturn` bundles.⁴

3.3 Navigation

The spacecraft pointing information present in SPICE C kernels is known to have errors on the order of dozens of NAC pixels. Before processing each image, an automatic navigation procedure (French et al. 2014b, 2016) was performed that matched the image with features predicted to be present in it, such as stars and the edge of the A ring. Once a mosaic was created from a set of images, the quality of the navigation was evaluated by eye and, in some cases, the navigation of some images was manually adjusted for the best visual result. The (automatically or manually) adjusted C matrix for each image is reported in a supplemental file called `IMGID_reproj_img_suppl.txt` in the `data_reproj_img` collection. See Section 4.2.1.

The outcome of that visual evaluation is recorded for each mosaic as the coded `nav_quality` field of the global index files (Section 4.4.1). No label contains an element of that name: the mosaic and reprojected image labels give the grade in words instead, in the `comment` element of their `Observation_Area`, in the form: the subjective quality of the navigation for all of the images for this mosaic is "good". The grade is subjective and describes the mosaic as a whole:

- **G** (good): the ring core runs continuously across the mosaic, with no visible radial step or tilt where one source image abuts the next.
- **F** (fair): small radial steps or tilts are visible at some image boundaries, but the core is unambiguous at every longitude and photometry is not noticeably affected.
- **P** (poor): at least one source image is misregistered enough to displace the core visibly. The mosaic remains usable for many purposes, but measurements at the affected longitudes should be treated with caution.

²https://pds-rings.seti.org/holdings/calibrated/COISS_2xxx/

³I/F is a dimensionless value that gives the ratio of the observed intensity to the incoming flux from the light source, assuming a perfect Lambert surface illuminated at normal incidence. See French et al. (2012) for more details.

⁴https://pds-rings.seti.org/pds4/bundles/cassini_iss

Because the grade covers the whole mosaic, a mosaic graded **F** or **P** may still contain a majority of well-navigated images.

3.4 Reprojection

Each calibrated image was converted to a reprojected image by mapping the pixels in the source image onto a two-dimensional grid with corotating longitude on the X axis and distance from the nominal F ring core (Δr) on the Y axis.

3.4.1 F ring orbit

Various orbital fits for the F ring core have been developed over time (Albers et al. 2012; Cooper et al. 2013; Cuzzi et al. 2024). We use the orbit fit from Albers et al. (2012), Table 3, fit #2, throughout:

$$\begin{aligned} n &= 581.964^\circ / \text{day} & a &= 140,221.3 \text{ km} & e &= 0.00235 \\ \varpi_0 &= 24.2^\circ & \dot{\varpi} &= 2.70025^\circ / \text{day} \\ \Omega_0 &= 15.0^\circ & \dot{\Omega} &= -2.68778^\circ / \text{day} \end{aligned}$$

The angles ϖ_0 and Ω_0 are the values at $ET = 0$, the J2000 epoch (2000-01-01T12:00:00 Barycentric Dynamical Time, TDB), where ET is the ephemeris time SPICE reports; the rates $\dot{\varpi}$ and $\dot{\Omega}$ carry them forward to the time of each observation, as in the core radius formula below. This epoch is separate from the corotation epoch defined below, which is used only to relate corotating and inertial longitude.

Inertial ring longitude is measured from the reference direction — the ascending node of Saturn's equatorial plane on the Earth's mean equator of J2000 — in the direction of orbital motion along that plane to the ring's ascending node, and thence along the ring plane.

Corotating longitude is based on a reference frame that rotates with the mean motion of the F ring. We use 2007-01-01T00:00:00Z as the epoch when corotating longitude and inertial longitude are equal. To convert from inertial to corotating longitude:

$$corot = \left(inertial - n \frac{ET - epoch_ET}{86400} \right) \bmod 360$$

and to convert from corotating to inertial longitude:

$$inertial = \left(corot + n \frac{ET - epoch_ET}{86400} \right) \bmod 360$$

where ET is the ephemeris (SPICE) time of the image in seconds and $epoch_ET$ is the corotating epoch in seconds, equal to 220881665.1839181.

Note that the choice of epoch is arbitrary, and when comparing absolute corotating longitudes from different sources it is important to convert them to a common reference frame. For example, Murray et al. (2008) used 2007-01-01T12:00:00Z and French et al. (2012) used 2004-01-01T12:00:00Z, while French et al. (2014a) used the same epoch adopted here.

To determine the radius of the F ring core, we use the standard formula:

$$r_{\text{core}} = \frac{a(1 - e^2)}{1 + e \cos(\lambda - (\varpi_0 + \dot{\varpi} \frac{ET}{86400}))}$$

where λ is the inertial longitude (converted from the corotating longitude as described above), so that the argument of the cosine is the true anomaly.

3.4.2 Reprojection process

The reprojection grid runs from 0° to 359.98° in steps of 0.02° (~ 49 km) for corotating longitude (18,000 pixels), and -1000 km to $+1000$ km in steps of 5 km for Δr (401 pixels). This gives an aspect ratio of roughly 10:1. For each corotating longitude, the radius is relative to the absolute radius of the core at the associated inertial longitude and time. This means that, while the core of the F ring varies in radius from 139,892 km to 140,551 km throughout its orbit, the core in the reprojected image will always look like an approximately straight line at $\Delta r = 0$, corresponding to $Y = 200$. Note that the odd number of pixels on the Y axis provides an equal number of radii on either side of the core, with the predicted core precisely in the center.

As the majority of reprojected images cover only a few degrees of corotating longitude, archiving files of the complete $401 \times 18,000$ grid would be extremely wasteful. As a result, the reprojected image data products are clipped on the X axis to the minimum and maximum corotating longitudes that contain valid data; see the associated label for information on these values. In the case where the extent of a reprojected image wraps around from 359.98° to 0° , the minimum corotating longitude is larger than the maximum corotating longitude, and the reprojected image is stored in two sections, first from the minimum corotating longitude to 359.98° and then from 0° to the maximum corotating longitude. See Section 4.2.1 for the formulas that map a longitude to a column in this case. In all cases, if there is missing data within the remaining grid (either due to insufficient radial extent available in the source image, or discontinuous corotating longitudes), it is marked with the sentinel value -999 .

During the reprojection process, several other parameters are computed for each corotating longitude, as the mean over that radial slice:

- Phase angle ($^\circ$)
- Emission angle ($^\circ$)
- Incidence angle ($^\circ$)
- Radial resolution (km/pixel)
- Longitudinal resolution ($^\circ$ /pixel)

Following standard planetary ring practice, incidence and emission angles are measured from the ring-plane normal on the sunlit side. The incidence angle is therefore always below 90° , and an emission angle below 90° means the ring is being viewed from its lit side.

These values are reported in a table `IMGID_reproj_img_metadata_params.tab` (Section 4.2.1) that is associated with each reprojected image. Other parameters describing the orbits of the F ring, Prometheus, and Pandora at the time of the source image are included as well:

- Core Radius (distance from Saturn in km, computed using the formula above)
- Longitude of Ascending Node ($^\circ$)
- Longitude of Pericenter ($^\circ$)
- True Anomaly ($^\circ$)
- Corotating Longitude of Prometheus ($^\circ$) in the same corotating frame used for the F ring
- Radius (distance from Saturn) of Prometheus (km)
- Corotating Longitude of Pandora ($^\circ$) in the same corotating frame used for the F ring

- Radius (distance from Saturn) of Pandora (km)

The positions of Prometheus and Pandora were computed with SPICE, taking the satellite positions from the Saturn system ephemeris `sat393.bsp`, planetary positions from `de438.bsp`, and the ring-plane orientation from the planetary constants kernel `cpck15Dec2017.tpc`. The metakernel archived in this bundle's `spice_kernels` collection (Section 4.5) lists every kernel used anywhere in the production of this bundle, which is a larger set than these three.

When the predicted position of Prometheus or Pandora falls inside a product's valid data, the moon is named as a target in that product's label and the label says either that "its presence has been visually confirmed" or that "its presence has not been visually confirmed". The confirmation means only this: someone looked at the mosaic and saw the moon in the pixels. A moon whose predicted position lands inside the valid longitudes but that was not picked out by eye — because the prediction falls on a pixel with no data, or outside the radial range the source images actually filled, or because the moon is too faint at that resolution — gets the second wording. The check is a visual one rather than a measurement, so no positional accuracy should be inferred from it. The label of an individual reprojected image never claims confirmation for itself, because the inspection was made against the mosaic rather than image by image.

Mean, minimum, and maximum values for many of these parameters are included in the label for each reprojected image as well. See Section 7.1 for a complete list of the metadata parameters provided.

3.5 Mosaic creation

Mosaics were created by stitching together one or more reprojected images. The resulting grid for a mosaic is the same $401 \times 18,000$ pixel grid as used for each reprojected image (Section 3.4). For each corotating longitude in the mosaic, the entire radial stripe is taken from a single image. The image is chosen first based on which image has the most valid pixels at that corotating longitude; when there is a tie, the image with the smallest mean radial resolution is used. This guarantees that the resulting mosaic will have the most data and the highest resolution possible, but does not guarantee that the selected reprojected images will change monotonically with corotating longitude. This can result in a slight temporal "blurring" across small ranges of corotating longitude.

As with reprojected images, several parameters are stored for each corotating longitude and provided in a table file `OBSID_mosaic_metadata_params.tab` (Section 4.2.3) associated with each mosaic:

- Source reprojected image number
- SPICE ET of source image
- Mean phase angle (°)
- Mean emission angle (°)
- Incidence angle (°)
- Mean radial resolution (km/pixel)
- Mean longitudinal resolution (°/pixel)
- Core Radius (distance from Saturn in km, computed using the formula above)
- Longitude of Ascending Node (°)
- Longitude of Pericenter (°)
- True Anomaly (°)
- Corotating Longitude of Prometheus (°) in the same corotating frame used for the F ring
- Radius (distance from Saturn) of Prometheus (km)

- Corotating Longitude of Pandora (°) in the same corotating frame used for the F ring
- Radius (distance from Saturn) of Pandora (km)

Mean, minimum, and maximum values for many of these parameters are included in the label for each mosaic as well. See Section 7.1 for a complete list of the metadata parameters provided.

3.6 Background modeling

The final step in the data production pipeline is modeling and subtracting the background gradient for each mosaic. The background gradient is modeled separately for each corotating longitude using a linear function. Regions normally consisting of the innermost 50 rows ($\Delta r = -1000$ to -755 km) and the outermost 50 rows ($\Delta r = +755$ to $+1000$ km) are used to fit a linear model, which is then subtracted from the complete 401-row radial slice. When creating the model, pixel values that are statistically anomalous compared to nearby longitudes are automatically masked. This removes stars, Prometheus and Pandora, and bad pixels from the background model computation. In some cases, the standard margins are not appropriate. Reasons include large objects such as Prometheus covering an entire margin, the reprojected source images not providing data for the entire radial range (which would require the background margins to be increased in size), and very low-resolution images where the F ring bleeds over into the background region (which would require background margins to be reduced). Such determinations were made through careful examination of each mosaic and the extent of the background margin was manually adjusted as needed. Whenever possible, the new margins were chosen to have minimal effect on computation of equivalent width.⁵ The chosen margins are given as the coded `bkgnd_lower_limit` and `bkgnd_upper_limit` fields of the global background-subtracted mosaic index (Section 4.4.1); the label of a background-subtracted mosaic has no elements of those names, describing the margins in words in the `comment` element of its `Observation_Area` instead, as in “available data from 910 to 1000 km closer to Saturn and 755 to 1000 km further from Saturn”. Both index fields are signed values of Δr in km and bound the ring itself, so the background is every row outside them: the interior region runs from -1000 km up to but not including `bkgnd_lower_limit` (normally -750 km), and the exterior region runs from just beyond `bkgnd_upper_limit` (normally $+750$ km) through $+1000$ km.

In cases where, even after adjusting the margins, fewer valid pixels than the minimum set for that mosaic remained in either background region at some longitudes once the anomalous pixels had been masked, those longitudes were marked as invalid and deleted from the background-subtracted mosaic; the label of each background-subtracted mosaic states the two minimums. As a result, the background-subtracted mosaic may have fewer (and sometimes substantially fewer) valid longitudes than the original mosaic. An example is shown in Figure 6.

The model is fit to noisy data, so subtracting it leaves the background regions scattered about zero. Roughly half of the pixels away from the ring in a background-subtracted mosaic are consequently negative. This is expected. Keep these values instead of clipping them at zero; discarding the negative half of the noise biases any integral over a radial slice, equivalent width included.

Each background-subtracted mosaic was inspected and the result recorded as the coded `bkgnd_quality` field of the global background-subtracted mosaic index (Section 4.4.1); as with `nav_quality`, the label states the grade in words in its `Observation_Area` `comment` rather than as an element of that name. Like `nav_quality` (Section 3.3), the grade is subjective:

⁵Equivalent width is a photometric value that integrates the brightness of a radial slice, taking into account the resolution of the image.

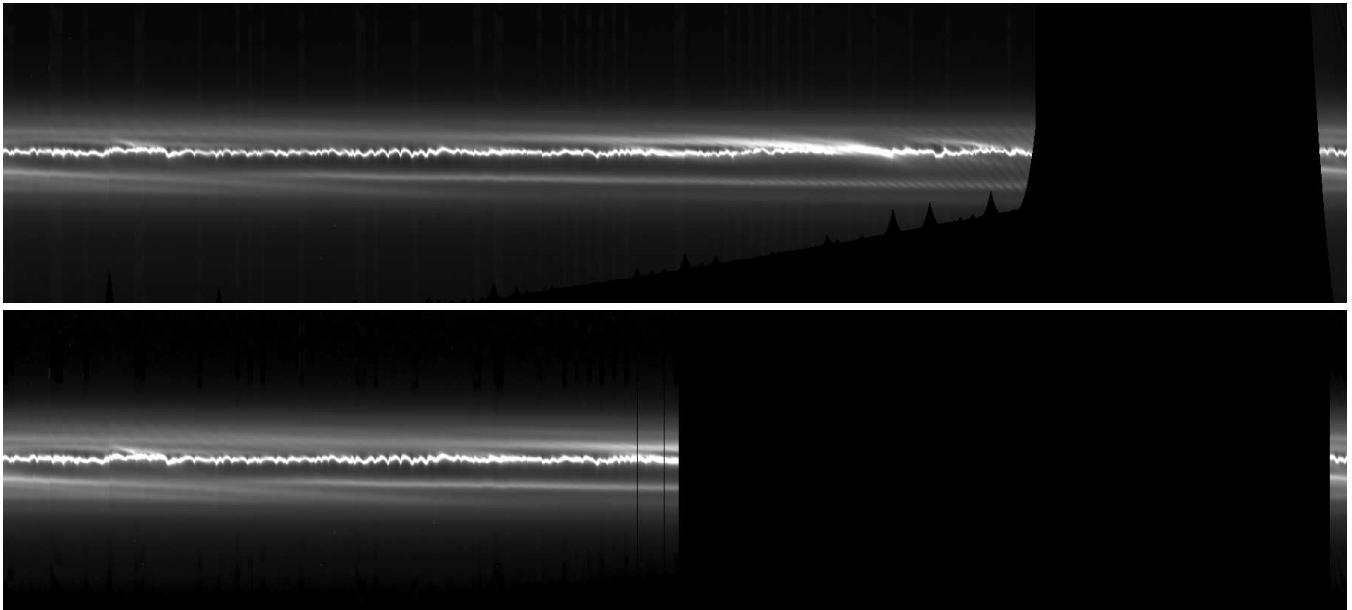


Figure 6: Original mosaic iss_007ri_hpmdfmov001_prime (top) and associated background-subtracted mosaic (bottom). The lack of some data in the original mosaic (to the lower left of the large blank area) prevents the creation of a background model for those longitudes, resulting in a background-subtracted mosaic with substantially fewer valid longitudes.

- **G** (good): the regions away from the ring are flat after subtraction at essentially all longitudes.
- **F** (fair): a residual gradient or offset remains at some longitudes, small enough that equivalent widths are still usable.
- **P** (poor): the model is visibly wrong over a substantial range of longitudes, usually because too little background data was available or because the F ring dust sheet extends into the background region.

4 Bundle organization and directory structure

The Cassini ISS F Ring Mosaics *bundle* is organized in a hierarchical directory structure, with each directory containing a PDS4 *collection* of a different type of *product*. There are data and browse collections for reprojected images, mosaics, and background-subtracted mosaics, plus collections for documentation, miscellaneous files (the global index files), context products, SPICE kernels, and XML schemas. The top-level directory structure is:

```
cassini_iss_fring_mosaics_rsfrench2025/
├── bundle.lblx
├── readme.txt
├── browse_mosaic/
├── browse_mosaic_bkg_sub/
├── browse_reproj_img/
├── context/
├── data_mosaic/
├── data_mosaic_bkg_sub/
├── data_reproj_img/
├── document/
├── miscellaneous/
├── spice_kernels/
└── xml_schema/
```

Each directory is described in the sections below. The `context` and `xml_schema` directories are used internally by PDS and will not be discussed further.

For reference, the following types of data products and support files are available:

- **Reprojected images:** These files contain rectangular reprojections of calibrated image files. Like the calibrated files, they are in units of I/F. Associated table files provide information about each longitude. These files are in the `data_reproj_img` directory. See Section 4.2.1.
- **Mosaics:** These files contain mosaics stitched together from one or more reprojected images. Like the reprojected images, they are in units of I/F. Associated table files provide information about each longitude. These files are in the `data_mosaic` directory. See Section 4.2.3.
- **Background-subtracted mosaics:** These files contain the same mosaics as above, but each mosaic has had a linear background model subtracted for each longitude. They are in units of I/F. Associated table files provide information about each longitude. These files are in the `data_mosaic_bkg_sub` directory. See Section 4.2.3.
- **Browse images:** Browse images in four sizes are available for each of the reprojected images, mosaics, and background-subtracted mosaics. These are *not* meant for scientific analysis and are only meant for visual examination of the contents of the data file. The original I/F data has been mapped to integers from 0 to 255 and non-linearly stretched to make dim features easier to see. The stretch uses a blackpoint at the minimum valid value or zero, whichever is greater, a whitepoint at the 99.8th-percentile value, and a gamma of 0.5. Because the blackpoint is never negative, negative values are all shown as black. These files are in the three `browse_*` directories. See Sections 4.2.2 and 4.2.4.
- **Documentation:** This *User Guide* is available in PDF format in the `document` directory. See Section 4.3.1.

- **Global index files:** These files provide a list of all reprojected images and mosaics in the bundle, along with tabulated metadata for each. They allow for easy programmatic searching on fields such as date and viewing geometry. These files are in the `miscellaneous` directory. See Section 4.4.1.
- **SPICE kernels:** A SPICE metakernel specifying the kernels that were used during navigation and data processing is provided in the `spice_kernels` directory. These files are not expected to be needed by most scientific users. See Section 4.5.
- **PDS4 support files:** These files are required by the PDS4 standard and contain information about the contents of the bundle directory. However, they are likely to be less useful to a scientist using the data in the bundle. These files include `bundle.lblx`, `collection_*.csv`, `collection_*.lblx`, and the contents of the `context` and `xml_schema` directories.

4.1 The top-level directory

The top-level directory contains the following files:

- `bundle.lblx`: The PDS4 bundle label, providing the high-level description of the entire archive.
- `readme.txt`: A text file directing the user to this *User Guide*.

4.2 The browse and data directories

For each derived data type — reprojected image (Sections 4.2.1 and 4.2.2), mosaic (Sections 4.2.3 and 4.2.4), and background-subtracted mosaic (Sections 4.2.3 and 4.2.4) — there is a `browse` and a `data` directory. These directories have the same internal structure:

```
<dirname>/
├── collection_<dirname>.csv
├── collection_<dirname>.lblx
├── OBSID1/
├── OBSID2/
├── ...
└── OBSIDn/
```

The `collection_<dirname>.csv` file contains a list of all of the data products present for all OBSIDs in this directory, and the accompanying label describes its metadata. The data products themselves are separated by OBSID for ease of retrieval and to keep the number of files in each subdirectory relatively small. Throughout this guide, OBSID is the Cassini observation name converted to lowercase, including any numeric chunk suffix (Section 3.1), and IMGID is the identifier of a single source image: its ten-digit Cassini spacecraft clock (SCLK) count followed by `n` for the NAC or `w` for the WAC, as in `1622049830n`. IMGID is the key that links a reprojected product back to the original COISS image.

Each derived data type has its own file structure within each OBSID directory.

4.2.1 data_reproj_img

This directory contains the individual reprojected images (Section 3.4) organized by OBSID. The directory structure is:

```
data_reproj_img/
├── collection_data_reproj_img.csv
```

```

├── collection_data_reproj_img.lblx
├── OBSID1/
│   ├── IMGID1_reproj_img.lblx
│   ├── IMGID1_reproj_img.img
│   ├── IMGID1_reproj_img_metadata_params.tab
│   ├── IMGID1_reproj_img_suppl.txt
│   └── ...
└── ...

```

The file `IMGID_reproj_img.lblx` contains all of the details about the reprojected image, including the reprojection and viewing geometry. Every excerpt in this section is taken from image 1622049830n of observation `iss_111rf_fmovie002_prime`:

```

<rings:epoch_reprojection_basis_utc>2007-01-01T00:00:00.000Z</rings:epoch_reprojection_basis_utc>
<rings:reprojection_plane>Equator</rings:reprojection_plane>
<rings:corotating_flag>Y</rings:corotating_flag>
<rings:corotation_rate unit="deg/s">0.006735694444</rings:corotation_rate> <!-- 581.964 deg/day -->
<rings:mean_phase_angle unit="deg">34.334</rings:mean_phase_angle>
<rings:minimum_phase_angle unit="deg">34.208</rings:minimum_phase_angle>
<rings:maximum_phase_angle unit="deg">34.373</rings:maximum_phase_angle>
<rings:mean_incidence_angle unit="deg">88.824</rings:mean_incidence_angle>
<rings:minimum_incidence_angle unit="deg">88.824</rings:minimum_incidence_angle>
<rings:maximum_incidence_angle unit="deg">88.824</rings:maximum_incidence_angle>
<rings:mean_emission_angle unit="deg">103.117</rings:mean_emission_angle>
<rings:minimum_emission_angle unit="deg">102.901</rings:minimum_emission_angle>
<rings:maximum_emission_angle unit="deg">103.339</rings:maximum_emission_angle>
<rings:minimum_corotating_ring_longitude unit="deg">186.60</rings:minimum_corotating_ring_longitude>
<rings:maximum_corotating_ring_longitude unit="deg">199.56</rings:maximum_corotating_ring_longitude>
<rings:minimum_ring_radius unit="km">138896.295</rings:minimum_ring_radius>
<rings:maximum_ring_radius unit="km">140916.702</rings:maximum_ring_radius>
<rings:Reprojection_Grid_Parameters>
  <local_identfier>reproj_grid</local_identfier>
  <rings:reprojection_grid_radial_sampling_interval
    unit="km">5.0</rings:reprojection_grid_radial_sampling_interval>
  <rings:reprojection_grid_longitudinal_sampling_interval
    unit="deg">0.02</rings:reprojection_grid_longitudinal_sampling_interval>
  <rings:mean_radial_resolution unit="km">5.853</rings:mean_radial_resolution>
  <rings:minimum_radial_resolution unit="km">5.613</rings:minimum_radial_resolution>
  <rings:maximum_radial_resolution unit="km">6.487</rings:maximum_radial_resolution>
  <rings:mean_longitudinal_resolution unit="deg">0.01007</rings:mean_longitudinal_resolution>
  <rings:minimum_longitudinal_resolution unit="deg">0.00963</rings:minimum_longitudinal_resolution>
  <rings:maximum_longitudinal_resolution unit="deg">0.01046</rings:maximum_longitudinal_resolution>
</rings:Reprojection_Grid_Parameters>

```

and the details of the image file, `IMGID_reproj_img.img`:

```

<File_Area_Observational>
  <File>
    <file_name>1622049830n_reproj_img.img</file_name>
    <creation_date_time>2026-09-28T21:45:52Z</creation_date_time>
    <file_size unit="byte">1040996</file_size>
    <md5_checksum>3d19ed500563eab78063a95eb57cbdd1</md5_checksum>
  </File>
  <!-- The calibrated, reprojected single image -->
  <Array_2D_Image>
    <local_identfier>reproj_image</local_identfier>
    <offset unit="byte">0</offset>
    <axes>2</axes>
    <axis_index_order>Last Index Fastest</axis_index_order>
    <description>
      Line is the radial axis and Sample is the corotating longitude axis.
      Line 0 is the innermost row, at a delta radius of -1000 km from the
      F ring core, and the delta radius increases outward by 5 km per line,
    </description>
  </Array_2D_Image>

```

```

so that delta radius = -1000 + 5 * Line km and the modeled core lies
at Line 200. Sample 0 is the minimum corotating longitude given by
rings:minimum_corotating_ring_longitude and the longitude increases
by 0.02 degrees per sample, wrapping through 360 degrees when the
minimum is greater than the maximum.
</description>
<Element_Array>
  <data_type>IEEE754LSBSingle</data_type>
  <unit>I/F</unit>
</Element_Array>
<Axis_Array>
  <axis_name>Line</axis_name> <!-- vertical (delta radius) -->
  <elements>401</elements>
  <sequence_number>1</sequence_number>
</Axis_Array>
<Axis_Array>
  <axis_name>Sample</axis_name> <!-- horizontal (corotating longitude) -->
  <elements>649</elements>
  <sequence_number>2</sequence_number>
</Axis_Array>
<Special_Constants>
  <!-- The data were missing or corrupted in the original image (either because it was off
the edge of the FOV or because of a transmission error resulting in a partial or
corrupted image) -->
  <missing_constant>-999</missing_constant>
</Special_Constants>
</Array_2D_Image>
</File_Area_Observational>

```

The image file is a pure binary file with no headers or record separators. Data is stored in LSB IEEE 754 single-precision (32-bit) floating-point format. The Y axis (Δr) always has size 401 (which covers the radial distance from the core of -1000 km to $+1000$ km in steps of 5 km). The array is stored with the Sample (longitude) index varying fastest — all longitudes at one radius, then the next radius — so that it reshapes directly to $(401, N)$ in row-major order. Line index 0 is the innermost row, at $\Delta r = -1000$ km, and Δr increases outward with the Line index, so that $\Delta r = -1000 + 5 \times \text{Line}$ km and the predicted core lies at Line 200. The X axis (corotating longitude) has a variable size depending on the number of longitudes that contain valid data (see Section 3.4). Any array elements that do not contain valid data will be marked with the value -999 . See Section 5 for information on how to read these files with software.

An associated metadata file, `IMGID_reproj_img_metadata_params.tab`, contains information about each *valid* longitude. This table has the general format:

```

rings:corotating_ring_longitude,rings:observed_event_tdb,rings:inertial_ring_longitude,
rings:radial_resolution,rings:longitudinal_resolution,rings:incidence_angle,rings:phase_angle,
rings:emission_angle,core_radius,longitude_ascending_node,longitude_pericenter,true_anomaly,
corotating_longitude_prometheus,radius_prometheus,corotating_longitude_pandora,radius_pandora
186.60, 296628214.587, 272.212, 6.487, 0.00963, 88.824, 34.369, 103.339, 139916.702, 147.322,
294.690, 337.522, 274.741, 139453.621, 68.056, 141137.545
186.62, 296628214.587, 272.232, 6.484, 0.00963, 88.824, 34.369, 103.339, 139916.658, 147.322,
294.690, 337.542, 274.741, 139453.621, 68.056, 141137.545
186.64, 296628214.587, 272.252, 6.480, 0.00964, 88.824, 34.368, 103.338, 139916.614, 147.322,
294.690, 337.562, 274.741, 139453.621, 68.056, 141137.545
186.66, 296628214.587, 272.272, 6.477, 0.00964, 88.824, 34.368, 103.337, 139916.571, 147.322,
294.690, 337.582, 274.741, 139453.621, 68.056, 141137.545
186.68, 296628214.587, 272.292, 6.473, 0.00964, 88.824, 34.367, 103.337, 139916.527, 147.322,
294.690, 337.602, 274.741, 139453.621, 68.056, 141137.545
[...]
```

where the rows are in ascending corotating-longitude order and cover only the longitudes that contain valid data. Because a reprojected image that wraps around at 360° is stored starting at its minimum corotating longitude and continuing through 360° to its maximum (see Section 3.4), and because the image may

retain sentinel-filled columns at longitudes with no valid data, the rows do *not* necessarily correspond one-to-one, in sequence, to the X indices of the image array; to associate a row with an image column, use the corotating ring longitude field instead of the row position. The column holding a given corotating longitude is $\text{col} = \text{round}(((\text{longitude} - \text{min_corotating_ring_longitude}) \bmod 360) / 0.02)$, and column c holds longitude $(\text{min_corotating_ring_longitude} + 0.02 \times c) \bmod 360$; the modulo handles the wraparound case (the example programs of Section 5.3.4 do this). This table gives the actual corotating longitude for each row as well as additional useful information such as the inertial longitude, radial resolution, angular (longitudinal) resolution, phase angle, emission angle, and various orbital parameters. See Sections 3.4 and 7 for details about these fields and Section 5 for information on how to read these files with software.

Finally, a supplemental information file, `IMGID_reproj_img_suppl.txt`, describes the navigation process and the derived pointing for the image (see Section 3.3):

```
This file contains a C-matrix that describes the rotation from the J2000 reference
frame to the camera pointing based upon analysis of the contents of the image.
The three rows of the matrix are the camera X, Y and Z axes expressed in J2000
coordinates, so the third row is the boresight and matches the RA and Dec below.
The roll is the angle of the camera X axis about the boresight, measured from the
direction of increasing right ascension and positive towards increasing
declination.
```

```
Source Data Product ID = 1622049830n_calib
Image Start Time (SCLK) = 1/1622049829.163
Image Start Time (UTC) = 2009-05-26T16:42:27.992Z
Image Mid Time (SCLK) = 1/1622049830.012
Image Mid Time (UTC) = 2009-05-26T16:42:28.402Z
Image Stop Time (SCLK) = 1/1622049830.118
Image Stop Time (UTC) = 2009-05-26T16:42:28.812Z
Trajectory Kernels Query Time = Observation mid time
Stellar Aberration Correction = No
Light Travel Time Correction = No
Navigation Type = Stars
Navigated Boresight RA = 142.804285 deg (09h31m13.028s)
Navigated Boresight Dec = -11.801585 deg (-011d48m05.706s)
Navigated Boresight Roll = -44.920922 deg
C-Matrix =
-0.3130222448  -0.6513481158  -0.6912038095
-0.5422429512  -0.4749368367   0.6931144083
-0.7797369147   0.5917606215  -0.2045231298
```

This pointing information is only provided for archival completeness and is not expected to be used by most scientific users.

4.2.2 *browse_reproj_img*

This directory contains browse images for each reprojected image. The directory structure is:

```
browse_reproj_img/
├── collection_browse_reproj_img.csv
├── collection_browse_reproj_img.lblx
├── OBSID1/
│   ├── IMGID1_browse_reproj_img.lblx
│   ├── IMGID1_browse_reproj_img_full.png
│   ├── IMGID1_browse_reproj_img_med.png
│   ├── IMGID1_browse_reproj_img_small.png
│   └── IMGID1_browse_reproj_img_thumb.png
```

```
└─┬─ ...
   │
   └─ ...
```

Four sizes of browse image are provided. All of them omit the longitudes where no data are available, so a discontinuous set of longitudes is drawn as adjacent, and the width they are derived from is the number of longitudes that hold data. Call that width V . “Full” images are V or 800 pixels wide, whichever is greater, by 401 pixels high. “Med” images are $V/10$ or 400 pixels wide, whichever is greater, by 400 pixels high. “Small” images are 200×200 and “Thumb” images are 100×100 (width $\times$ height). All four are produced by resampling to those dimensions; the one case in which that leaves the original pixel grid untouched is a Full image already at least 800 pixels wide. Because most reprojected images are narrower than that, a typical Full image is stretched horizontally to reach the 800-pixel minimum and a typical Med image is 400 pixels wide. The Med image carries the observation name and the image name, separated by a slash, drawn in the upper left corner above the words `reproj img`; the Small and Thumb images carry the image name alone, for example `1622049830n` above `reproj img`; the Full image is drawn without any text. All sizes are contrast-stretched to use the full available grayscale and are not calibrated for scientific use. Radial pixels without valid data, within the longitudes that are kept, are drawn in black. The rows run with Δr increasing upward, following the Bottom to Top display direction declared in the data label: the top row of each browse image is the outer edge of the grid ($\Delta r = +1000$ km) and the bottom row is the inner edge ($\Delta r = -1000$ km), which is the reverse of the order in which the rows are stored in the image array (Section 4.2.1).

4.2.3 *data_mosaic and data_mosaic_bkg_sub*

These directories have identical structure and contain the original and background-subtracted mosaics, respectively. The directory structures are:

```
data_mosaic/
├─ collection_data_mosaic.csv
├─ collection_data_mosaic.lblx
├─ OBSID1/
│   ├── OBSID1_mosaic.lblx
│   ├── OBSID1_mosaic.img
│   ├── OBSID1_mosaic_metadata_params.tab
│   ├── OBSID1_mosaic_metadata_src_imgs.tab
│   └─ ...
└─ ...
```

and:

```
data_mosaic_bkg_sub/
├─ collection_data_mosaic_bkg_sub.csv
├─ collection_data_mosaic_bkg_sub.lblx
├─ OBSID1/
│   ├── OBSID1_mosaic_bkg_sub.lblx
│   ├── OBSID1_mosaic_bkg_sub.img
│   ├── OBSID1_mosaic_bkg_sub_metadata_params.tab
│   ├── OBSID1_mosaic_bkg_sub_metadata_src_imgs.tab
│   └─ ...
└─ ...
```

The format of the `OBSID1_mosaic[_bkg_sub].lblx` and `OBSID1_mosaic[_bkg_sub].img` files is very similar to the corresponding `reproj_img` files described earlier. The only difference is that the mosaic image files are always $401 \times 18,000$, since mosaic images are never clipped to include only valid data (see Sections 3.5 and 4.2.1).

The `OBSID1_mosaic[_bkg_sub].metadata_params.tab` file is also similar to the corresponding `reproj_img` file, with the exception that it contains one additional column:

```
rings:corotating_ring_longitude,image_index,rings:observed_event_tdb,rings:inertial_ring_longitude,
rings:radial_resolution,rings:longitudinal_resolution,rings:incidence_angle,rings:phase_angle,
rings:emission_angle,core_radius,longitude_ascending_node,longitude_pericenter,true_anomaly,
corotating_longitude_prometheus,radius_prometheus,corotating_longitude_pandora,radius_pandora
[...]
4.60,      8, 296602047.773, 273.960,      5.289,  0.03542,  88.824,  38.053,  93.478, 139911.390,
      148.136, 293.872, 340.088, 273.625, 139319.212,  70.524, 142262.216
4.62,      8, 296602047.773, 273.980,      5.286,  0.03542,  88.824,  38.053,  93.477, 139911.351,
      148.136, 293.872, 340.108, 273.625, 139319.212,  70.524, 142262.216
4.64,      8, 296602047.773, 274.000,      5.282,  0.03543,  88.824,  38.053,  93.477, 139911.312,
      148.136, 293.872, 340.128, 273.625, 139319.212,  70.524, 142262.216
4.66,      8, 296602047.773, 274.020,      5.279,  0.03543,  88.824,  38.053,  93.477, 139911.273,
      148.136, 293.872, 340.148, 273.625, 139319.212,  70.524, 142262.216
4.68,      8, 296602047.773, 274.040,      5.276,  0.03543,  88.824,  38.053,  93.477, 139911.234,
      148.136, 293.872, 340.168, 273.625, 139319.212,  70.524, 142262.216
[...]
```

The same caution about row position applies here, but the mapping is simpler. A mosaic is never clipped, so its 18,000 columns always span the full grid: column c holds corotating longitude $0.02 \times c$ degrees, and the column holding a given longitude is $\text{col} = \text{round}(\text{longitude} / 0.02)$. The rows of the table, however, cover only the longitudes that contain valid data, so row i of the table is generally not column i of the image.

The new column is `image_index`, which indicates which reprojected image was used to provide data for this particular corotating longitude. The number in the `image_index` column references the file `OBSID1_mosaic[_bkg_sub].metadata_src_imgs.tab`, which contains a list of the LIDVIDs of the reprojected images that were used to create this mosaic:

```
image_index,LIDVID
0, urn:nasa:pds:cassini_iss_fring_mosaics_rsfrench2025:data_reproj_img:1622022571n_reproj_img::1.0
1, urn:nasa:pds:cassini_iss_fring_mosaics_rsfrench2025:data_reproj_img:1622022704n_reproj_img::1.0
2, urn:nasa:pds:cassini_iss_fring_mosaics_rsfrench2025:data_reproj_img:1622022841n_reproj_img::1.0
[...]
```

The source-image table of a background-subtracted mosaic lists only the images that still supply at least one longitude after the background subtraction, numbered from 0 in the same order, so where the subtraction removed every longitude of some image the two products of one observation assign different `image_index` values to the same image (seven observations in this bundle; for example the background-subtracted table of `iss_007ri_hpmrdmov001_prime` lists 105 of the mosaic's 169 images). To relate the two products, match the LIDVIDs in the two source-image tables, or match rows on `rings:corotating_ring_longitude`; do not match on `image_index`.

While all of these files are in fixed-width format for rapid reading with appropriate software, they can also be read as if they were CSV files by standard spreadsheet software. In this case you must specify that the file is delimited by commas.

See Sections 3.4, 3.5, and 7 for details about these fields and Section 5 for information on how to read these files with software.

4.2.4 *browse_mosaic* and *browse_mosaic_bkg_sub*

These directories contain browse images for the corresponding mosaics. Browse images are provided in four sizes:

- Thumb: 100×100 pixels
- Small: 200×200 pixels
- Med: 1800×400 pixels
- Full: $18,000 \times 401$ pixels

Every size other than Full is resampled from the mosaic; the sizes above are width $\times$ height. The Med image carries the full observation name drawn in the upper left corner, above the word `mosaic` (or `bkgnd-sub mosaic` for a background-subtracted mosaic). The Small and Thumb images carry an abbreviated form of the observation name instead, with the leading instrument code and the trailing prime-instrument code dropped, so that `iss_111rf_fmovie002_prime` appears as `111rf_fmovie002` and chunk 1 of `iss_112rf_fmovie002_prime_1` appears as `112rf_fmovie002/1`. The Full image is drawn without any text. Each browse image is contrast-stretched to use the full available grayscale and is not calibrated for scientific use. Missing data in the mosaic is drawn in black. The rows run with Δr increasing upward, following the Bottom to Top display direction declared in the data label: the top row of each browse image is the outer edge of the grid ($\Delta r = +1000$ km) and the bottom row is the inner edge ($\Delta r = -1000$ km), which is the reverse of the order in which the rows are stored in the mosaic array (Section 4.2.3). A background-subtracted mosaic omits every longitude at which no background model could be fit (Section 3.6), so its browse images can show more black columns than those of the original mosaic of the same observation.

4.3 The document directory

This directory contains the document collection and has the structure:

```
document/
├── collection_document.csv
├── collection_document.lblx
├── user_guide/
│   ├── display_reproj_img.py
│   ├── f-ring-mosaics-user-guide.lblx
│   ├── f-ring-mosaics-user-guide.pdf
│   ├── find_prometheus_closest_approaches.py
│   ├── mosaic_utils.py
│   ├── plot_ews_df.py
│   └── plot_ews_ma.py
```

4.3.1 *user_guide*

The `user_guide` directory contains the *User Guide* (this document) and example programs.

4.4 The miscellaneous directory: the global index files

This directory contains the global index files and has the structure:

```
miscellaneous/
```

```

|— collection_miscellaneous.csv
|— collection_miscellaneous.lblx
|— global_mosaic_bkg_sub_index.lblx
|— global_mosaic_bkg_sub_index.tab
|— global_mosaic_index.lblx
|— global_mosaic_index.tab
|— global_reproj_img_index.lblx
|— global_reproj_img_index.tab

```

4.4.1 The global index files

The `miscellaneous` directory contains three global index files, one for reprojected images, one for mosaics, and one for background-subtracted mosaics. Each contains summary information about every product in its collection. The provided information is extensive and is designed to be useful to researchers who want to search for images or mosaics meeting certain requirements.

The following notes may be present for each reprojected image and mosaic. The `notes` column of an index file holds only the codes themselves; if more than one applies they are separated by a semicolon. The sentence that goes with a code is spelled out in prose in the `comment` element of the `Observation_Area` of the mosaic label — a PDS4 label element, not an XML comment. Items in braces below are replaced by the observation name or by a computed value in that prose.

- M1: Because Cassini observed the same inertial longitudes for more than one orbit of the F ring during {root_obsid}, we have split the observation into multiple chunks, each corresponding to one complete orbit of the F ring. This mosaic consists of {root_obsid} chunk {chunk}. Other mosaics are available in this bundle for {root_obsid} with different date ranges representing the other available observation chunks.
- M2: Because observation {root_obsid} consists of two distinct "movies" covering approximately the same corotating longitudes but taken at inertial longitudes roughly 180 degrees apart, we have split the observation into two chunks. This mosaic consists of {root_obsid} chunk {chunk}. The other mosaic is available as {other_obsid}.
- M3: Because observation {root_obsid} consists of multiple "movies" covering approximately the same corotating longitudes but taken at different inertial longitudes (not 180 degrees apart), we have split the observation into multiple chunks, with the same two closing sentences as M1.
- M4: Because Cassini observed multiple distinct inertial and corotating longitudes during {root_obsid}, each making its own "movie", we have split the observation into multiple chunks, with the same two closing sentences as M1.
- B: In a background-subtracted mosaic label, this background-subtracted mosaic contains substantially fewer valid longitudes than the original mosaic due to insufficient data being available to create a background model; the label of the mosaic itself carries the same note the other way round, as the background-subtracted version of this mosaic contains substantially fewer valid longitudes than this mosaic, because insufficient data were available to create a background model at the missing longitudes.
- C: Some source images contained corrupted or missing data that could not be repaired during the construction of the mosaic.
- E: Some source images may have been overexposed and the data values clipped; use caution when using this mosaic for photometry.

- O: The sequence of source images used to create this mosaic was designed to observe a stellar occultation of the F ring core. As such, a star is present in each source image and may appear in the mosaic multiple times depending on how the reprojected images were stitched together. In addition, the source images were taken at roughly the same corotating longitudes and thus have significant overlap in the mosaic. To fully explore the occultation, use the reprojected images.
- R: During this time, Cassini followed one corotating longitude for {total_secs} seconds ({total_hours} hours) by observing multiple inertial longitudes covering the (possibly discontinuous) {diff_inertial} degrees from {min_inertial} to {max_inertial}.
- N: During this time, Cassini observed multiple corotating longitudes at multiple inertial longitudes for {total_secs} seconds ({total_hours} hours). The inertial longitudes covered the (possibly discontinuous) {diff_inertial} degrees from {min_inertial} to {max_inertial}.

4.5 The `spice_kernels` directory

This directory contains the SPICE kernel collection, which consists of a single SPICE metakernel `kernels.ker` listing all kernels used to process the data for this bundle (see Section 6.3). As with the supplemental files described in Section 4.2.1, these files are only provided for archival completeness and are not expected to be used by most scientific users.

5 Reading labels and data product files

5.1 Introduction

The files in this bundle are archived in PDS4 format. PDS4 has the following basic structure:

- A *bundle* consists of one or more *collections*
- A *collection* consists of one or more *data products*
- Each *data product* is defined by a PDS4 label (`.lblx`), which gives information about the data and also includes information about the additional associated files that contain the actual data. These data files may be binary or text and may be stored in a variety of formats.

A full discussion of the PDS4 format is beyond the scope of this document, and readers are referred to the resources in Section 6.1 for additional reading. However, we will endeavor to provide enough information here to allow reading of these various files for research purposes without comprehensive knowledge of the PDS4 standard.

Two types of software can be used to read PDS4 labels and data files: generic off-the-shelf software that does not understand PDS4 concepts, and specific tools or programming libraries that understand PDS4 labels. We will start by discussing the former.

5.2 Using off-the-shelf programs

Label files (`.lblx`), table files (`.tab`), and CSV files (`.csv`) are standard text files that can be read with any text editor. However, while this is sufficient to give you an overview of the contents of the file, each file has a specific format designed to be read and parsed by specific types of programs, and this can be done without the need for PDS4-specific software. Using the correct program in the correct way will make using these files much easier and more productive.

5.2.1 Label files

Label files are written in XML (eXtensible Markup Language), a hierarchical file format much like a nested outline. A text editor that understands XML gives the best results. Such an editor will likely have two main features:

- The ability to collapse and expand different levels of the hierarchy, so as to hide parts of the label that are not needed for the task at hand; and
- Syntax coloring, where the open and close tags (`<tagname>data</tagname>`) are color-coded for ease of identification.

As text editors are constantly coming in and out of favor and are platform-specific, we will not recommend specific tools here. However, almost every sophisticated editor or programming IDE (Integrated Development Environment) either natively supports XML or has an XML plugin. In most cases the editor will automatically recognize the file as being in XML format, but in some cases it might be necessary to explicitly invoke the XML mode if the editor relies on the file extension (since `.lblx` is not a normal XML extension).

5.2.2 Table files

Files with the extension `.tab` (including the metadata parameters, source image index, and global indexes) are formatted as *fixed-width comma-delimited tables*. That is, they look like:

```
Value123    ,ValueA  ,ValueB
Value45678  ,ValueCDE,ValueF
```

This format allows software that knows the width of each column to read the table efficiently (Section 5.3), while also letting the file be read by the wide range of software that supports comma-separated value (CSV) files, such as spreadsheet programs. When opening the table file, a spreadsheet program may identify it as fixed-width; if it does, it will append the comma to each field. To properly read a table in a spreadsheet program, be sure to specify that the file is “delimited” and that the delimiter is “;”; also specify that you want whitespace stripped if possible.

5.2.3 CSV files

Files with the extension `.csv` (mainly the collection files) are formatted as *variable-width comma-separated value* files. These files can be read the same way as the above table files, except it is more likely programs will correctly identify them as delimited rather than fixed-width from the beginning.

5.2.4 Image files

Files with the extension `.img` hold the reprojected images and mosaics. They are raw binary arrays with no header, no record separators, and no padding: LSB (little-endian) IEEE 754 single-precision floats, with the longitude index varying fastest so that the data reshapes directly to 401 radii by N longitudes, and with the value `−999` marking any element that has no valid data (Section 4.2.1). The example programs described in Section 5.3.4 demonstrate how to read and display these files.

5.3 Using PDS4-specific software

A variety of software is available for reading PDS4-format data. Note that over time tools evolve and are replaced, and the URLs or package names where tools can be found change. This section includes tools and their locations at the time of publication. To find new or updated tools, use the resources in Section 6.1.

5.3.1 The Small Bodies Node tools

The PDS Small Bodies Node (SBN) has a variety of software useful for reading PDS4-format data. This includes:

- **PDS4 Viewer:** This standalone application is capable of reading PDS4 labels and associated tabular and binary data files and displaying them in various formats.
https://pds-smallbodies.astro.umd.edu/tools/pds4_tools_info/pds4_viewer.html
- **Python PDS4 Tools:** This Python module will read and display PDS4 labels and associated tabular and binary data files. Data read this way can be used directly in Python programs for further analysis.
https://pdssbn.astro.umd.edu/tools/pds4_tools_info/python_pds4_tools.html
- **ReadPDS:** This IDL module will read and display PDS4 labels and associated tabular and binary data files. Data read this way can be used directly in IDL programs for further analysis.
https://pdssbn.astro.umd.edu/tools/tools_readPDS.shtml

5.3.2 The Ring-Moon Systems Node tools

The PDS Ring-Moon Systems Node (RMS) provides a Python PDS4 table reader that can also fetch tables from remote locations. It is based on the SBN PDS4 label reader.

<https://rms-pdstable.readthedocs.io/en/latest/>

5.3.3 The Planetary Data Reader

The Planetary Data Reader (PDR) is a Python package that aims to read all PDS3 and PDS4 format data through a uniform interface. PDS4 support is based on the SBN PDS4 label reader.

<https://pdr.readthedocs.io/en/latest/>

5.3.4 Example programs

The F ring bundle includes example Python programs explicitly designed for reading and manipulating the mosaic and reprojected image data products. Each program is thoroughly commented to aid the user in learning the concepts involved. The programs can be found in the `document/user_guide` directory (Section 4.3.1). They require Python 3 with `pds4_tools` (Section 5.3.1), `numpy` and `matplotlib` installed; the functions and programs that return metadata as a `DataFrame` also require `pandas`, which is imported only when one of those is used.

- `mosaic_utils.py`: A support module that is used by all other example programs and that can be directly imported into user-developed programs. It supplies the following functions:
 - `get_element`: Search a label for a particular element name and return its value.
 - `contrast_stretch`: Non-linearly stretch an image based on blackpoint, whitepoint, and gamma values.
 - `read_mosaic_ma` and `read_reproj_img_ma`: Read an F ring mosaic or reprojected image into a numpy masked array with the metadata returned as a structured numpy array.
 - `read_mosaic_ma_df` and `read_reproj_img_ma_df`: Read an F ring mosaic or reprojected image into a numpy masked array with the metadata returned as a pandas `DataFrame`.
 - `get_mosaic_name_from_lid` and `get_reproj_img_name_from_lid`: Convert a LID to the name of the mosaic or reprojected image.
 - `get_mosaic_name_from_mosaic_label` and `get_reproj_img_name_from_label`: Get the name of the mosaic from the mosaic label or the name of a reprojected image from the reprojected image label.
 - `get_mosaic_name_from_reproj_img_label`: Get the name of the mosaic that incorporates a reprojected image from the reprojected image label.
 - `read_index_np`: Read a global index into a numpy array.
 - `read_index_df`: Read a global index into a pandas `DataFrame`.
- `display_reproj_img.py`: Display a single reprojected image.
- `plot_ews_ma.py`: Plot a mosaic along with its equivalent width profile using numpy masked arrays for computation.
- `plot_ews_df.py`: Plot a mosaic along with its equivalent width profile using pandas `DataFrames` for computation (same functionality as the previous program, but the internals are different for illustration).

- `find_prometheus_closest_approaches.py`: Search the global reprojected image index file for images where Prometheus is visible and display the 10 images in which Prometheus is closest to the F ring core.

Here are example uses:

```
$ python display_reproj_img.py \
    /path/to/bundle/data_reproj_img/iss_036rf_fmovie001_vims/
    1545564642n_reproj_img.lblx
```

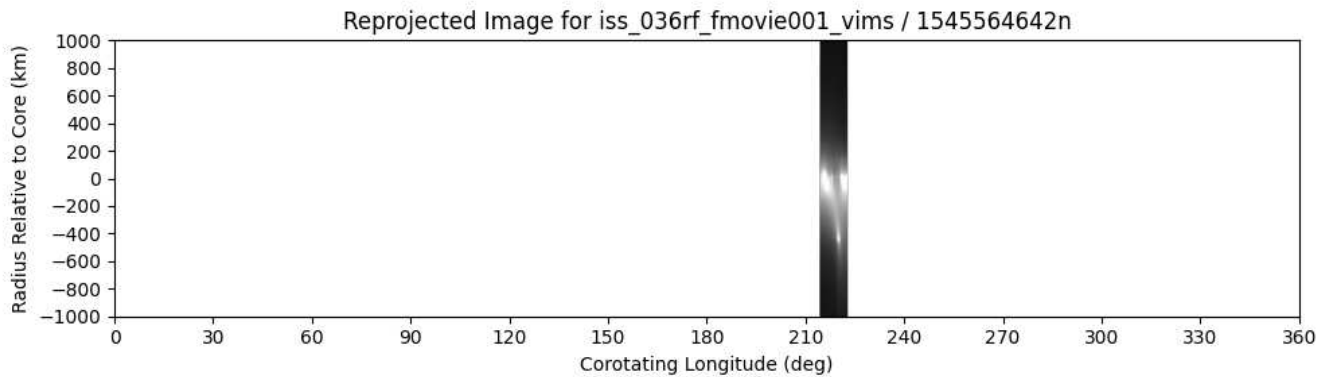


Figure 7: Output of `display_reproj_img.py` for reprojected image 1545564642n of observation `iss_036rf_fmovie001_vims`.

```
$ python plot_ews_ma.py \
    /path/to/bundle/data_mosaic_bkg_sub/iss_036rf_fmovie001_vims/
    iss_036rf_fmovie001_vims_mosaic_bkg_sub.lblx
```

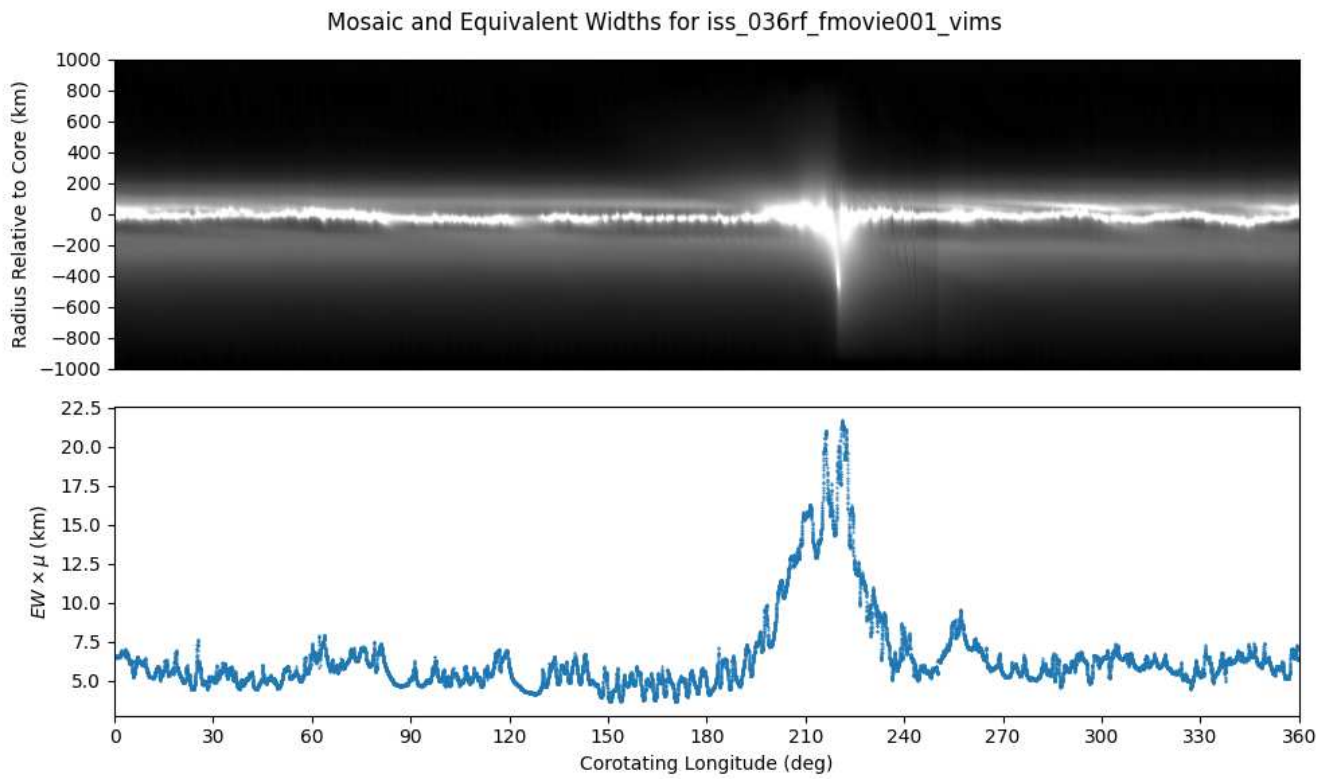


Figure 8: Output of `plot_ews_ma.py` for the background-subtracted mosaic of `iss_036rf_fmovie001_vims`: the mosaic above its equivalent width profile.

```
$ python find_prometheus_closest_approaches.py /path/to/bundle
```

Ten closest Prometheus approaches

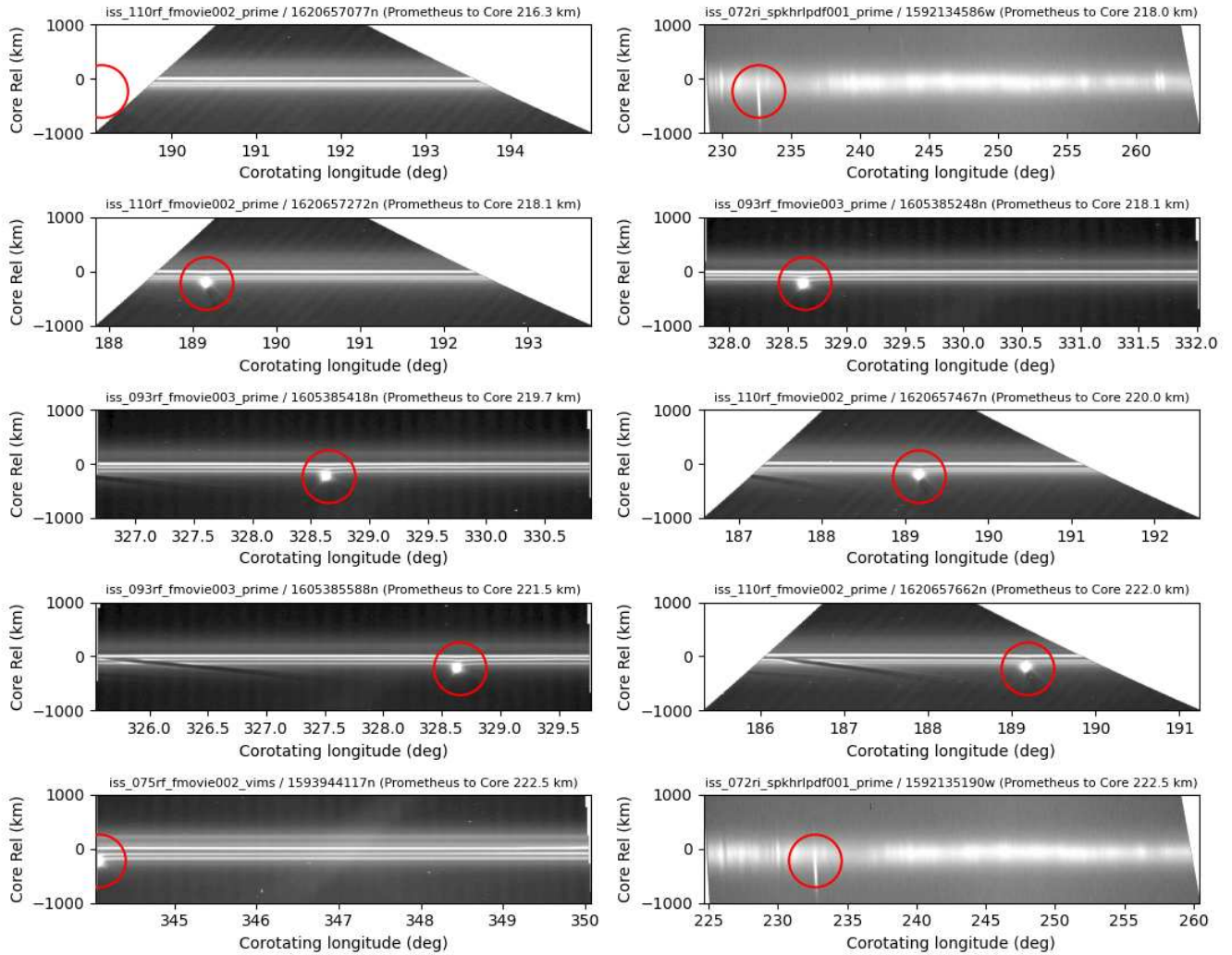


Figure 9: Output of `find_prometheus_closest_approaches.py`: the ten reprojected images in which Prometheus comes closest to the F ring core.

5.3.5 F ring bundle analysis programs

In addition to the example programs included with the archived bundle, further software is available to aid researchers in viewing and analyzing the data. As this software evolves rapidly, it is not included with the bundle, but is instead available in this GitHub repository:

<https://github.com/SETI/fring-mosaics-bundle-software>

6 External references and further reading

This section provides a curated list of external references, documentation, and further reading relevant to the Cassini ISS F Ring Mosaics bundle.

6.1 PDS4 standards and documentation

- NASA Planetary Data System (PDS) Home: <https://pds.nasa.gov/>
- PDS4 Documents: <https://pds.nasa.gov/datastandards/documents/>
- PDS4 Information Model Specification: <https://pds.nasa.gov/datastandards/documents/im/>
- PDS4 Data Dictionaries: <https://pds.nasa.gov/datastandards/dictionaries/>

6.2 Cassini mission and instrumentation

- Cassini-Huygens Mission Overview: <https://science.nasa.gov/mission/cassini/>
- Cassini Imaging Science Subsystem (ISS) Overview: https://pds-atmospheres.nmsu.edu/data_and_services/atmospheres_data/Cassini/inst-iss.html
- Cassini Imaging Science Subsystem (ISS) Data User's Guide (PDS3): <https://doi.org/10.17189/1504135>
- Cassini ISS Narrow Angle Camera (NAC) Details: https://pds-rings.seti.org/pds4/bundles/cassini_iss_saturn/document/iss-na-camera-description.txt
- Cassini ISS Wide Angle Camera (WAC) Details: https://pds-rings.seti.org/pds4/bundles/cassini_iss_saturn/document/iss-wa-camera-description.txt

6.3 SPICE toolkit

- SPICE Toolkit Documentation: <https://naif.jpl.nasa.gov/naif/documentation.html>
- SPICE Tutorials and Reference: <https://naif.jpl.nasa.gov/naif/tutorials.html>

6.4 Additional resources

- PDS Ring-Moon Systems Node: <https://pds-rings.seti.org/>
- PDS Small Bodies Node: <https://pdssbn.astro.umd.edu/>
- PDS Cartography and Imaging Sciences Node: <https://pds-imaging.jpl.nasa.gov/>

6.5 References

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7 Metadata and global index file fields

7.1 Reprojected image and mosaic metadata params

Units are uniform throughout this section and throughout the bundle. All angles, longitudes, and true anomalies are in degrees; all radii and radial distances are in km; radial resolution is in km/pixel and longitudinal resolution in °/pixel; coverage is given in percent; and all times given as `tdb` (including `rings:observed_event_tdb`) are ephemeris times in seconds past the J2000 epoch, 2000-01-01T12:00:00 Barycentric Dynamical Time (TDB) — the same quantity SPICE calls *ET*. Times given as `date_time` are Coordinated Universal Time (UTC).

The following fields are tabulated for each valid corotating longitude in each reprojected image (`IMGID_reproj_img_metadata_params.tab`), mosaic (`OBSID_mosaic_metadata_params.tab`), and background-subtracted mosaic (`OBSID_mosaic_bkg_sub_metadata_params.tab`).

Table 3: Metadata parameter fields provided for each valid corotating longitude.

Field	Description
<code>rings:corotating_ring_longitude</code>	The corotating longitude for which values are being provided
<code>image_index</code>	(Mosaics only) The number of the reprojected image (from <code>OBSID_mosaic_metadata_src_imgs.tab</code> or <code>OBSID_mosaic_bkg_sub_metadata_src_imgs.tab</code>) that provided data for this corotating longitude
<code>rings:observed_event_tdb</code>	The mid-time of the source image that provided data for this corotating longitude (constant for one image)
<code>rings:inertial_ring_longitude</code>	The inertial longitude of the part of the source image that provided data for this corotating longitude
<code>rings:radial_resolution</code>	The radial resolution of the part of the source image that provided data for this corotating longitude
<code>rings:longitudinal_resolution</code>	The angular longitudinal resolution of the part of the source image that provided data for this corotating longitude
<code>rings:incidence_angle</code>	The incidence angle of the ring plane at the time of the source image (constant for one image)
<code>rings:phase_angle</code>	The phase angle of the part of the source image that provided data for this corotating longitude
<code>rings:emission_angle</code>	The emission angle of the part of the source image that provided data for this corotating longitude
<code>core_radius</code>	The radius of the ideal F ring core from the orbit model at this corotating longitude
<code>longitude_ascending_node</code>	The longitude of the ascending node (relative to J2000) from the orbit model at the time of the source image (constant for one image)

Field	Description
longitude_pericenter	The longitude of pericenter (relative to J2000) from the orbit model at the time of the source image (constant for one image)
true_anomaly	The true anomaly from the orbit model corresponding to this corotating longitude
corotating_longitude_prometheus	The corotating longitude of Prometheus at the time of the source image, in the same reference frame used for the F ring core (constant for one image)
radius_prometheus	The orbital radius (distance from Saturn center) of Prometheus at the time of the source image (constant for one image)
corotating_longitude_pandora	The corotating longitude of Pandora at the time of the source image, in the same reference frame used for the F ring core (constant for one image)
radius_pandora	The orbital radius (distance from Saturn center) of Pandora at the time of the source image (constant for one image)

7.2 Reprojected image global index

The following fields appear in `global_reproj_img_index.tab`.

Table 4: Fields in the reprojected image global index.

Field	Description
pds:logical_identifier	The PDS4 logical identifier, <i>e.g.</i> urn:nasa:pds:cassini_iss_fring_mosaics_rsfrench2025:data_reproj_img:1874525875w_reproj_img
cassini:observation_id	The Cassini observation name of the source image, <i>e.g.</i> IOSIC_276RB_COMPLITB3001_SI
file_spec	The path to the reprojected image label file relative to the bundle root, <i>e.g.</i> data_reproj_img/iosic_276rb_complitb3001_si/1874525875w_reproj_img.lblx
pds:start_date_time	The start time of the source image exposure, <i>e.g.</i> 2017-05-26T20:29:09Z
pds:stop_date_time	The stop time of the source image exposure
cassini:spacecraft_clock_start_count	The starting spacecraft clock count of the source image without the partition number (which is always 1/)
cassini:spacecraft_clock_stop_count	The ending spacecraft clock count of the source image without the partition number (which is always 1/)

Field	Description
num_valid_longitudes	The number of discrete longitudes (radial slices) that contain valid data; each radial slice represents 0.02° of corotating longitude
percent_coverage	The percentage of 360° containing valid data; computed as $\text{num_valid_longitudes} / 18,000 \times 100$
rings:minimum_corotating_ring_longitude	The minimum corotating longitude, representing the left edge of the reprojected image
rings:maximum_corotating_ring_longitude	The maximum corotating longitude, representing the right edge of the reprojected image; note that it is possible for min > max, in which case the corotating range wraps around at 360°
rings:minimum_inertial_ring_longitude	The minimum inertial longitude from the source image used to create the reprojected image
rings:maximum_inertial_ring_longitude	The maximum inertial longitude from the source image used to create the reprojected image; because this is a single exposure, the corotating and inertial longitude ranges have the same width
rings:mean_phase_angle	The mean (across all valid longitudes) phase angle
rings:minimum_phase_angle	The minimum (across all valid longitudes) phase angle
rings:maximum_phase_angle	The maximum (across all valid longitudes) phase angle
rings:mean_incidence_angle	The ring plane incidence angle at the time of the source image. The incidence angle does not vary with longitude and changes only slowly with time, so a single value is computed for the whole product; rings:minimum_incidence_angle and rings:maximum_incidence_angle are present in the label and carry that same value
rings:mean_emission_angle	The mean (across all valid longitudes) emission angle; the angle is < 90° on the lit side of the rings
rings:minimum_emission_angle	The minimum (across all valid longitudes) emission angle
rings:maximum_emission_angle	The maximum (across all valid longitudes) emission angle
rings:mean_radial_resolution	The mean (across all valid longitudes) radial resolution in km/pixel; radial resolution is the resolution in the radial direction of the source image used to populate each reprojected image pixel
rings:minimum_radial_resolution	The minimum (across all valid longitudes) radial resolution in km/pixel
rings:maximum_radial_resolution	The maximum (across all valid longitudes) radial resolution in km/pixel

Field	Description
rings:mean_longitudinal_resolution	The mean (across all valid longitudes) angular longitudinal resolution in °/pixel; angular longitudinal resolution is the resolution in the longitude direction (expressed as an angle) of the source image used to populate each reprojected image pixel
rings:minimum_longitudinal_resolution	The minimum (across all valid longitudes) angular longitudinal resolution in °/pixel
rings:maximum_longitudinal_resolution	The maximum (across all valid longitudes) angular longitudinal resolution in °/pixel
rings:minimum_ring_radius	The inner edge of the reprojection grid: the smallest (across all valid longitudes) ring radius (distance from Saturn center) the grid covers, which is <code>minimum_core_radius</code> – 1000 km
rings:maximum_ring_radius	The outer edge of the reprojection grid: the largest (across all valid longitudes) ring radius (distance from Saturn center) the grid covers, which is <code>maximum_core_radius</code> + 1000 km
mean_core_radius	The mean (across all valid longitudes) ring radius (distance from Saturn center) of the ideal F ring core from the orbit model (Section 3.4.1)
minimum_core_radius	The minimum (across all valid longitudes) ring radius (distance from Saturn center) of the F ring core from the orbit model
maximum_core_radius	The maximum (across all valid longitudes) ring radius (distance from Saturn center) of the F ring core from the orbit model
longitude_ascending_node	The longitude of the ascending node of the F ring (relative to J2000) from the orbit model at the time of the source image (Section 3.4.1)
longitude_pericenter	The longitude of pericenter of the F ring (relative to J2000) from the orbit model at the time of the source image (Section 3.4.1)
minimum_true_anomaly	The minimum (across all valid longitudes) true anomaly from the orbit model (Section 3.4.1)
maximum_true_anomaly	The maximum (across all valid longitudes) true anomaly from the orbit model (Section 3.4.1)
corotating_longitude_prometheus	The corotating longitude of Prometheus at the time of the source image in the same reference frame used for the F ring core
radius_prometheus	The orbital radius (distance from Saturn center) of Prometheus at the time of the source image
corotating_longitude_pandora	The corotating longitude of Pandora at the time of the source image in the same reference frame used for the F ring core

Field	Description
radius_pandora	The orbital radius (distance from Saturn center) of Pandora at the time of the source image
pds:creation_date_time	The date and time the reprojected image product was created
nav_quality	The navigation grade of the mosaic belonging to this image's observation. It is the overall grade for that group of images, not an assessment of this one image, and every reprojected image of the same observation carries the same value — including those that did not contribute to the mosaic (Section 3.1.6). Values are G = Good, F = Fair, P = Poor (Section 3.3)
notes	Notes about the mosaic belonging to this image's observation (Section 4.4.1)

7.3 Mosaic and background-subtracted mosaic global indexes

The following fields appear in `global_mosaic_index.tab` and `global_mosaic_bkg_sub_index.tab`. Rows marked “background-subtracted mosaic only” are present only in the latter.

Table 5: Fields in the mosaic and background-subtracted mosaic global indexes.

Field	Description
pds:logical_identifier	The PDS4 logical identifier, <i>e.g.</i> urn:nasa:pds:cassini_iss_fring_mosaics_rsfrench2025:data_mosaic:iosic_276rb_complitb3001_si_mosaic
cassini:observation_id	The Cassini observation name of the source images, <i>e.g.</i> IOSIC_276RB_COMPLITB3001_SI
file_spec	The path to the mosaic label file relative to the bundle root, <i>e.g.</i> data_mosaic/iosic_276rb_complitb3001_si/iosic_276rb_complitb3001_si_mosaic.lblx
pds:start_date_time	The start time of the earliest source image exposure, <i>e.g.</i> 2017-05-26T20:29:09Z
pds:stop_date_time	The stop time of the latest source image exposure
cassini:spacecraft_clock_start_count	The starting spacecraft clock count of the earliest source image without the partition number (which is always 1/)
cassini:spacecraft_clock_stop_count	The ending spacecraft clock count of the latest source image without the partition number (which is always 1/)
num_valid_longitudes	The number of discrete longitudes (radial slices) that contain valid data; each radial slice represents 0.02° of corotating longitude

Field	Description
percent_coverage	The percentage of 360° containing valid data; computed as $\text{num_valid_longitudes} / 18,000 \times 100$
rings:minimum_corotating_ring_longitude	The lower end of the range of corotating longitude that contains valid data. This is not the same quantity as the rings:minimum_corotating_ring_longitude of the product label, which always reports the full grid, 0.00° to 359.98°, because a mosaic is always written out to its full extent
rings:maximum_corotating_ring_longitude	The upper end of the range of corotating longitude that contains valid data, and likewise not the same quantity as the rings:maximum_corotating_ring_longitude of the product label; when the range of valid data wraps through 360° this value is less than the minimum
rings:minimum_inertial_ring_longitude	The minimum inertial longitude from the source images used to create the mosaic
rings:maximum_inertial_ring_longitude	The maximum inertial longitude from the source images used to create the mosaic
rings:mean_phase_angle	The mean (across all valid longitudes) phase angle
rings:minimum_phase_angle	The minimum (across all valid longitudes) phase angle
rings:maximum_phase_angle	The maximum (across all valid longitudes) phase angle
rings:mean_incidence_angle	The ring plane incidence angle during the times the source images were taken. The incidence angle does not vary with longitude and changes only slowly with time, so a single value is computed for the whole mosaic; rings:minimum_incidence_angle and rings:maximum_incidence_angle are present in the label and carry that same value
rings:mean_emission_angle	The mean (across all valid longitudes) emission angle; the angle is < 90° on the lit side of the rings
rings:minimum_emission_angle	The minimum (across all valid longitudes) emission angle
rings:maximum_emission_angle	The maximum (across all valid longitudes) emission angle
rings:mean_radial_resolution	The mean (across all valid longitudes) radial resolution in km/pixel; radial resolution is the resolution in the radial direction of the source image used to populate each mosaic pixel
rings:minimum_radial_resolution	The minimum (across all valid longitudes) radial resolution in km/pixel
rings:maximum_radial_resolution	The maximum (across all valid longitudes) radial resolution in km/pixel

Field	Description
rings:mean_longitudinal_resolution	The mean (across all valid longitudes) angular longitudinal resolution in °/pixel; angular longitudinal resolution is the resolution in the longitude direction (expressed as an angle) of the source image used to populate each mosaic pixel
rings:minimum_longitudinal_resolution	The minimum (across all valid longitudes) angular longitudinal resolution in °/pixel
rings:maximum_longitudinal_resolution	The maximum (across all valid longitudes) angular longitudinal resolution in °/pixel
rings:minimum_ring_radius	The inner edge of the reprojection grid: the smallest (across all valid longitudes) ring radius (distance from Saturn center) the grid covers, which is minimum_core_radius – 1000 km
rings:maximum_ring_radius	The outer edge of the reprojection grid: the largest (across all valid longitudes) ring radius (distance from Saturn center) the grid covers, which is maximum_core_radius + 1000 km
mean_core_radius	The mean (across all valid longitudes) ring radius (distance from Saturn center) of the ideal F ring core (Section 3.4.1)
minimum_core_radius	The minimum (across all valid longitudes) ring radius (distance from Saturn center) of the F ring core
maximum_core_radius	The maximum (across all valid longitudes) ring radius (distance from Saturn center) of the F ring core
mean_longitude_ascending_node	The mean longitude of the ascending node of the F ring (relative to J2000) from the orbit model across all of the source images (Section 3.4.1)
minimum_longitude_ascending_node	The minimum longitude of the ascending node of the F ring (relative to J2000) from the orbit model across all of the source images (Section 3.4.1)
maximum_longitude_ascending_node	The maximum longitude of the ascending node of the F ring (relative to J2000) from the orbit model across all of the source images (Section 3.4.1)
mean_longitude_pericenter	The mean longitude of pericenter of the F ring (relative to J2000) from the orbit model across all of the source images (Section 3.4.1)
minimum_longitude_pericenter	The minimum longitude of pericenter of the F ring (relative to J2000) from the orbit model across all of the source images (Section 3.4.1)
maximum_longitude_pericenter	The maximum longitude of pericenter of the F ring (relative to J2000) from the orbit model across all of the source images (Section 3.4.1)
minimum_true_anomaly	The minimum (across all valid longitudes) true anomaly from the orbit model (Section 3.4.1)

Field	Description
maximum_true_anomaly	The maximum (across all valid longitudes) true anomaly from the orbit model (Section 3.4.1)
mean_corotating_longitude_prometheus	The mean corotating longitude of Prometheus in the same reference frame used for the F ring core across all source images
minimum_corotating_longitude_prometheus	The minimum corotating longitude of Prometheus in the same reference frame used for the F ring core across all source images
maximum_corotating_longitude_prometheus	The maximum corotating longitude of Prometheus in the same reference frame used for the F ring core across all source images
mean_radius_prometheus	The mean orbital radius (distance from Saturn center) of Prometheus across all source images
minimum_radius_prometheus	The minimum orbital radius (distance from Saturn center) of Prometheus across all source images
maximum_radius_prometheus	The maximum orbital radius (distance from Saturn center) of Prometheus across all source images
mean_corotating_longitude_pandora	The mean corotating longitude of Pandora in the same reference frame used for the F ring core across all source images
minimum_corotating_longitude_pandora	The minimum corotating longitude of Pandora in the same reference frame used for the F ring core across all source images
maximum_corotating_longitude_pandora	The maximum corotating longitude of Pandora in the same reference frame used for the F ring core across all source images
mean_radius_pandora	The mean orbital radius (distance from Saturn center) of Pandora across all source images
minimum_radius_pandora	The minimum orbital radius (distance from Saturn center) of Pandora across all source images
maximum_radius_pandora	The maximum orbital radius (distance from Saturn center) of Pandora across all source images
pds:creation_date_time	The date and time the mosaic product was created
nav_quality	The overall quality of the navigation across all source images; values are G = Good, F = Fair, P = Poor (Section 3.3)
bkgnd_quality	(Background-subtracted mosaic only) The overall quality of the background model used for this mosaic; values are G = Good, F = Fair, P = Poor (Section 3.6)
notes	Notes about the mosaic (Section 4.4.1)
num_images	The number of reprojected images used to create this mosaic

Field	Description
min_image_name	The name of the earliest source image used in this mosaic
max_image_name	The name of the latest source image used in this mosaic
bkgnd_lower_limit	(Background-subtracted mosaic only) The inner edge of the ring region, as a signed Δr in km; the rows from $\Delta r = -1000$ km up to but not including this value are used for background modeling
bkgnd_upper_limit	(Background-subtracted mosaic only) The outer edge of the ring region, as a signed Δr in km; the rows beyond this value, up through $\Delta r = +1000$ km, are used for background modeling